CHAPTER 1

Geometric Camera Models

There are many types of imaging devices, from animal eyes to video cameras and radio telescopes, and they may or may not be equipped with lenses. For example, the first models of the *camera obscura* (literally, dark chamber) invented in the sixteenth century did not have lenses, but instead used a *pinhole* to focus light rays onto a wall or translucent plate and demonstrate the laws of perspective discovered a century earlier by Brunelleschi. Pinholes were replaced by more and more sophisticated lenses as early as 1550, and the modern photographic or digital camera is essentially a camera obscura capable of recording the amount of light striking every small area of its backplane (Figure 1.1).

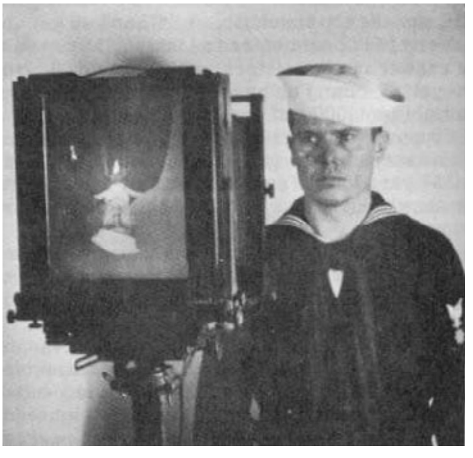

FIGURE 1.1: Image formation on the backplate of a photographic camera. *Figure from US NAVY MANUAL OF BASIC OPTICS AND OPTICAL INSTRUMENTS, prepared by the Bureau of Naval Personnel, reprinted by Dover Publications, Inc. (1969).*

The imaging surface of a camera is in general a rectangle, but the shape of the human retina is much closer to a spherical surface, and panoramic cameras may be equipped with cylindrical retinas. Imaging sensors have other characteristics. They may record a spatially discrete picture (like our eyes with their rods and cones, 35mm cameras with their grain, and digital cameras with their rectangular picture elements, or pixels), or a continuous one (in the case of old-fashioned TV tubes, for example). The signal that an imaging sensor records at a point on its retina may itself be discrete or continuous, and it may consist of a single number (as for a black-and-white camera), a few values (e.g., the RGB intensities for a color camera or the responses of the three types of cones for the human eye), many numbers (e.g., the responses of hyperspectral sensors), or even a continuous function of wavelength (which is essentially the case for spectrometers). Chapter 2 considers cameras as *radiometric* devices for measuring light energy, brightness, and color. Here, we focus instead on purely geometric camera characteristics. After introducing several models of image formation in Section 1.1—including a brief description of this process in the human eye in Section 1.1.4—we define the *intrinsic* and *extrinsic* geometric parameters characterizing a camera in Section 1.2, and finally show how to estimate these parameters from image data—a process known as *geometric camera calibration*—in Section 1.3.

1.1 IMAGE FORMATION

1.1.1 Pinhole Perspective

Imagine taking a box, using a pin to prick a small hole in the center of one of its sides, and then replacing the opposite side with a translucent plate. If you hold that box in front of you in a dimly lit room, with the pinhole facing some light source, say a candle, an inverted image of the candle will appear on the translucent plate (Figure 1.2). This image is formed by light rays issued from the scene facing the box. If the pinhole were really reduced to a point (which is physically impossible, of course), exactly one light ray would pass through each point in the plane of the plate (or *image plane*), the pinhole, and some scene point.

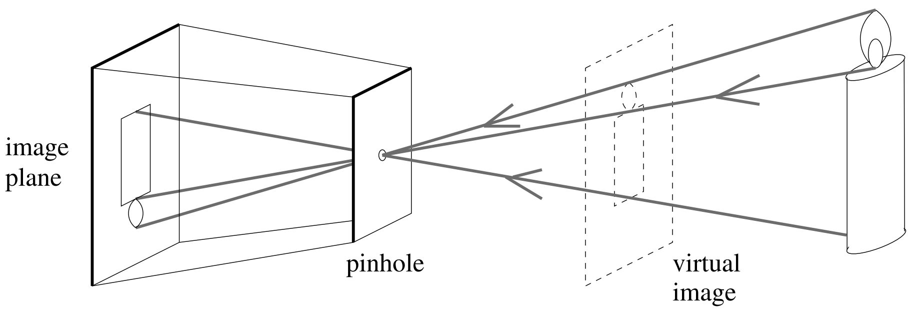

FIGURE 1.2: The pinhole imaging model.

In reality, the pinhole will have a finite (albeit small) size, and each point in the image plane will collect light from a cone of rays subtending a finite solid angle, so this idealized and extremely simple model of the imaging geometry will not strictly apply. In addition, real cameras are normally equipped with lenses, which further complicates things. Still, the *pinhole perspective* (also called *central perspective*) projection model, first proposed by Brunelleschi at the beginning of the fifteenth century, is mathematically convenient and, despite its simplicity, it often provides an acceptable approximation of the imaging process. Perspective projection creates inverted images, and it is sometimes convenient to consider instead a *virtual image* associated with a plane lying *in front* of the pinhole, at the same distance from it as the actual image plane (Figure 1.2). This virtual image is not inverted but is otherwise strictly equivalent to the actual one. Depending on the context, it may be more convenient to think about one or the other. Figure 1.3 (a) illustrates an obvious effect of perspective projection: the apparent size of objects depends on their distance. For example, the images b and c of the posts B and C have the same height, but A and C are really half the size of B. Figure 1.3 (b) illustrates another well-known effect: the projections of two parallel lines lying in some plane Φ appear to converge on a horizon line h formed by the intersection of the image plane Π with the plane parallel to Φ and passing through the pinhole. Note that the line L parallel to Π in Φ has no image at all.

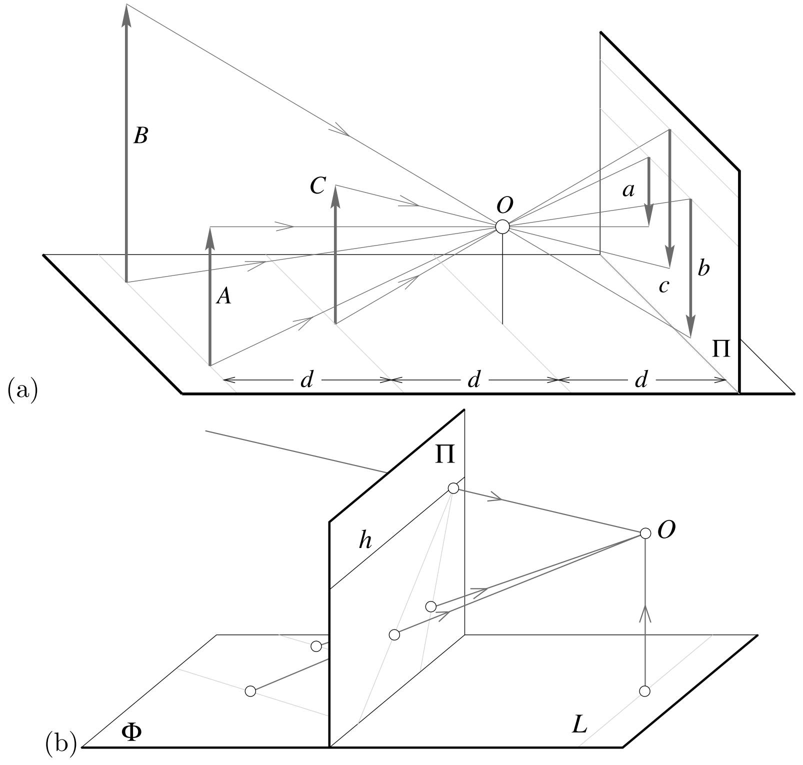

FIGURE 1.3: Perspective effects: (a) far objects appear smaller than close ones: The distance d from the pinhole O to the plane containing C is half the distance from O to the plane containing A and B; (b) the images of parallel lines intersect at the horizon (after Hilbert and Cohn-Vossen, 1952, Figure 127). Note that the image plane Π is *behind* the pinhole in (a) (physical retina), and *in front* of it in (b) (virtual image plane). Most of the diagrams in this chapter and in the rest of this book will feature the physical image plane, but a virtual one will also be used when appropriate, as in (b).

These properties are easy to prove in a purely geometric fashion. As usual, however, it is often convenient (if not quite as elegant) to reason in terms of reference frames, coordinates, and equations. Consider, for example, a coordinate system $(O, \boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k})$ attached to a pinhole camera, whose origin O coincides with the pinhole, and vectors $\boldsymbol{i}$ and $\boldsymbol{j}$ form a basis for a vector plane parallel to the image plane Π, itself located at a positive distance d from the pinhole along the vector $\boldsymbol{k}$ (Figure 1.4). The line perpendicular to Π and passing through the pinhole is called the optical axis, and the point c where it pierces Π is called the *image center*. This point can be used as the origin of an image plane coordinate frame, and it plays an important role in camera calibration procedures.

Let P denote a scene point with coordinates (X, Y, Z) and p denote its image

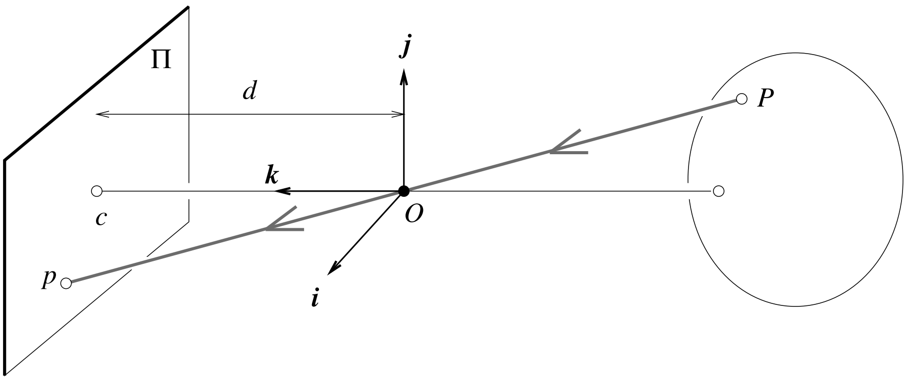

FIGURE 1.4: The perspective projection equations are derived in this section from the collinearity of the point P, its image p, and the pinhole O.

with coordinates (x, y, z). (Throughout this chapter, we will use uppercase letters to denotes points in space, and lowercase letters to denote their image projections.) Since p lies in the image plane, we have $z = d$. Since the three points P, O, and p are collinear, we have $\overrightarrow{Op} = \lambda \overrightarrow{OP}$ for some number λ, so

$$\begin{cases} x = \lambda X \\ y = \lambda Y \\ d = \lambda Z \end{cases} \iff \lambda = \frac{x}{X} = \frac{y}{Y} = \frac{d}{Z},$$

and therefore

$$\begin{cases} x = d\dfrac{X}{Z}, \\[2ex] y = d\dfrac{Y}{Z}. \end{cases} \tag{1.1}$$

1.1.2 Weak Perspective

As noted in the previous section, pinhole perspective is only an approximation of the geometry of the imaging process. This section discusses a coarser approximation, called *weak perspective*, which is also useful on occasion.

Consider the *fronto-parallel plane* Π_0 defined by $Z = Z_0$ (Figure 1.5). For any point P in Π_0 we can rewrite Eq. (1.1) as

$$\begin{cases} x = -mX, \\ y = -mY, \end{cases} \quad \text{where} \quad m = -\frac{d}{Z_0}. \tag{1.2}$$

Physical constraints impose that Z_0 be negative (the plane must be in front of the pinhole), so the *magnification* m associated with the plane Π_0 is positive. This name is justified by the following remark: consider two points P and Q in Π_0 and their images p and q (Figure 1.5); obviously, the vectors $\overrightarrow{PQ}$ and $\overrightarrow{pq}$ are parallel, and we have $||\overrightarrow{pq}|| = m||\overrightarrow{PQ}||$. This is the dependence of image size on object distance noted earlier.

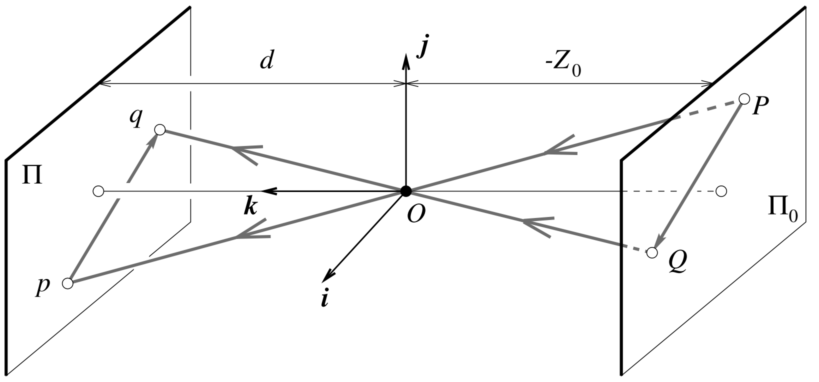

FIGURE 1.5: Weak-perspective projection. All line segments in the plane Π_0 are projected with the same magnification.

When a scene's relief is small relative to its average distance from the camera, the magnification can be taken to be constant. This projection model is called *weak perspective*, or *scaled orthography*.

When it is a priori known that the camera will always remain at a roughly constant distance from the scene, we can go further and normalize the image coordinates so that $m = -1$. This is *orthographic projection*, defined by

$$\begin{cases} x = X, \\ y = Y, \end{cases} \tag{1.3}$$

with all light rays parallel to the $\boldsymbol{k}$ axis and orthogonal to the image plane π (Figure 1.6). Although weak-perspective projection is an acceptable model for many imaging conditions, assuming pure orthographic projection is usually unrealistic.

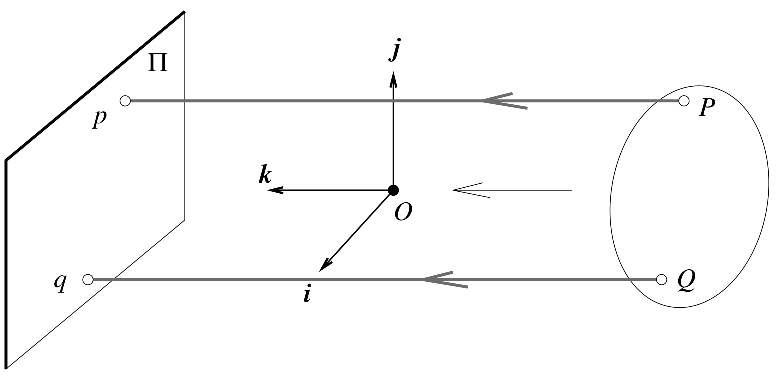

FIGURE 1.6: Orthographic projection. Unlike other geometric models of the image formation process, orthographic projection does not involve a reversal of image features. Accordingly, the magnification is taken to be negative, which is a bit unnatural but simplifies the projection equations.

1.1.3 Cameras with Lenses

Most real cameras are equipped with lenses. There are two main reasons for this: The first one is to gather light, because a single ray of light would otherwise reach each point in the image plane under ideal pinhole projection. Real pinholes have a finite size, of course, so each point in the image plane is illuminated by a cone of light rays subtending a finite solid angle. The larger the hole, the wider the cone and the brighter the image, but a large pinhole gives blurry pictures. Shrinking the pinhole produces sharper images but reduces the amount of light reaching the image plane, and may introduce *diffraction* effects. Keeping the picture in sharp focus while gathering light from a large area is the second main reason for using a lens.

Ignoring diffraction, interferences, and other physical optics phenomena, the behavior of lenses is dictated by the laws of geometric optics (Figure 1.7): (1) light travels in straight lines (*light rays*) in homogeneous media; (2) when a ray is reflected from a surface, this ray, its reflection, and the surface normal are coplanar, and the angles between the normal and the two rays are complementary; and (3) when a ray passes from one medium to another, it is *refracted*, i.e., its direction changes. According to Snell's law, if r_1 is the ray incident to the interface between two transparent materials with indices of refraction n_1 and n_2, and r_2 is the refracted ray, then r_1, r_2, and the normal to the interface are coplanar, and the angles α_1 and α_2 between the normal and the two rays are related by

$$n_1 \sin \alpha_1 = n_2 \sin \alpha_2. \tag{1.4}$$

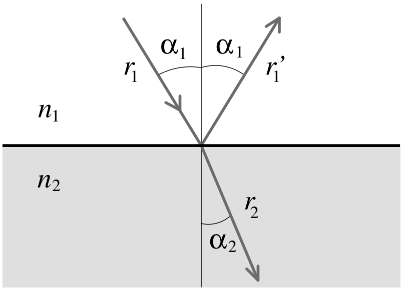

FIGURE 1.7: Reflection and refraction at the interface between two homogeneous media with indices of refraction n_1 and n_2.

In this chapter, we will only consider the effects of refraction and ignore those of reflection. In other words, we will concentrate on lenses as opposed to *catadioptric optical systems* (e.g., telescopes) that may include both reflective (mirrors) and refractive elements. Tracing light rays as they travel through a lens is simpler when the angles between these rays and the refracting surfaces of the lens are assumed to be small, which is the domain of *paraxial* (or *first-order*) geometric optics, and Snell's law becomes $n_1\alpha_1 \approx n_2\alpha_2$. Let us also assume that the lens is rotationally symmetric about a straight line, called its *optical axis*, and that all refractive surfaces are spherical. The symmetry of this setup allows us to determine

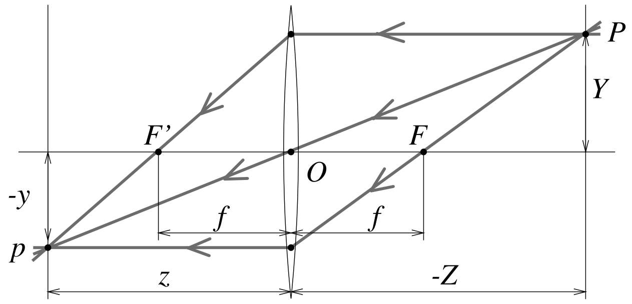

FIGURE 1.8: A thin lens. Rays passing through O are not refracted. Rays parallel to the optical axis are focused on the focal point F'.

the projection geometry by considering lenses with circular boundaries lying in a plane that contains the optical axis. In particular, consider a lens with two spherical surfaces of radius R and index of refraction n. We will assume that this lens is surrounded by vacuum (or, to an excellent approximation, by air), with an index of refraction equal to 1, and that it is *thin*, i.e., that a ray entering the lens and refracted at its right boundary is immediately refracted again at the left boundary.

Consider a point P located at (negative) depth Z off the optical axis, and denote by (PO) the ray passing through this point and the center O of the lens (Figure 1.8). It easily follows from the paraxial form of Snell's law that (PO) is not refracted, and that all the other rays passing through P are focused by the thin lens on the point p with depth z along (PO) such that

$$\frac{1}{z} - \frac{1}{Z} = \frac{1}{f}, \tag{1.5}$$

where $f = \frac{R}{2(n-1)}$ is the *focal length* of the lens.

Note that the equations relating the positions of P and p are exactly the same as under pinhole perspective projection if we take $d = z$ since P and p lie on a ray passing through the center of the lens, but that points located at a distance $-Z$ from O will be in sharp focus only when the image plane is located at a distance z from O on the other side of the lens that satisfies Eq. (1.5), the *thin lens equation*. Letting $Z \to -\infty$ shows that f is the distance between the center of the lens and the plane where objects such as stars (that are effectively located at $Z = -\infty$) focus. The two points F and F' located at distance f from the lens center on the optical axis are called the *focal points* of the lens. In practice, objects within some range of distances (called *depth of field* or *depth of focus*) will be in acceptable focus. As shown in the problems at the end of this chapter, the depth of field increases with the *f-number* of the lens, i.e., the ratio between the focal length of the lens and its diameter.

Note that the *field of view* of a camera, i.e., the portion of scene space that actually projects onto the retina of the camera, is not defined by the focal length alone but also depends on the effective area of the retina (e.g., the area of film that can be exposed in a photographic camera, or the area of the sensor in a digital camera; see Figure 1.9).

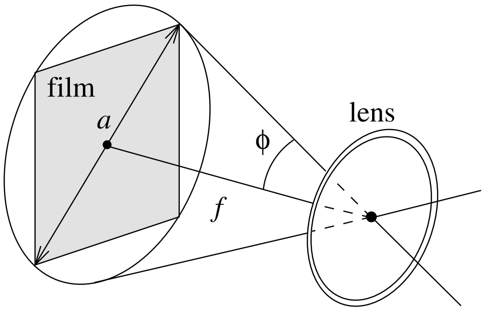

FIGURE 1.9: The field of view of a camera. It can be defined as 2ϕ, where $\phi \stackrel{\text{def}}{=} \arctan \frac{a}{2f}$, a is the diameter of the sensor (film, CCD, or CMOS chip), and f is the focal length of the camera.

A more realistic model of simple optical systems is the *thick lens.* The equations describing its behavior are easily derived from the paraxial refraction equation, and they are the same as the pinhole perspective and thin lens projection equations, except for an offset (Figure 1.10). If H and H' denote the *principal points* of the lens, then Eq. (1.5) holds when $-Z$ (resp. z) is the distance between P (resp. p) and the plane perpendicular to the optical axis and passing through H (resp. H'). In this case, the only undeflected ray is along the optical axis.

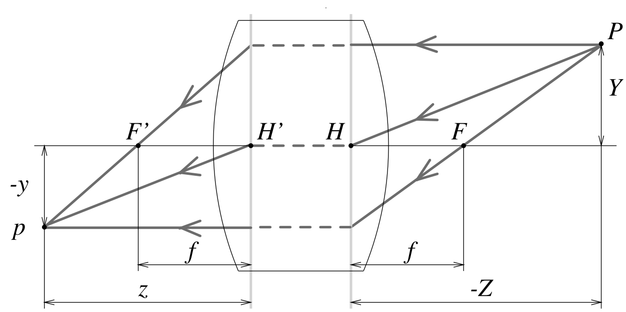

FIGURE 1.10: A simple thick lens with two spherical surfaces.

Simple lenses suffer from a number of *aberrations.* To understand why, let us remember that the paraxial refraction model is only an approximation, valid when the angle α between each ray along the optical path and the optical axis of the length is small and $\sin \alpha \approx \alpha$. This corresponds to a first-order Taylor expansion of the sine function. For larger angles, additional terms yield a better approximation, and it is easy to show that rays striking the interface farther from the optical axis are focused closer to the interface. The same phenomenon occurs for a lens, and it is the source of two types of *spherical aberrations* (Figure 1.11 [a]): Consider a point P on the optical axis and its paraxial image p. The distance between p and the intersection of the optical axis with a ray issued from P and refracted by the lens is called the longitudinal spherical aberration of that ray. Note that if an image plane Π were erected in P, the ray would intersect this plane at some distance from the axis, called the transverse spherical aberration of that ray. Together, all rays passing through P and refracted by the lens form a circle of confusion centered in P as they intersect Π. The size of that circle will change if we move Π along the optical axis. The circle with minimum diameter is called the *circle of least confusion*, and its center does not coincide (in general) with p.

(a)

(b)

(c)

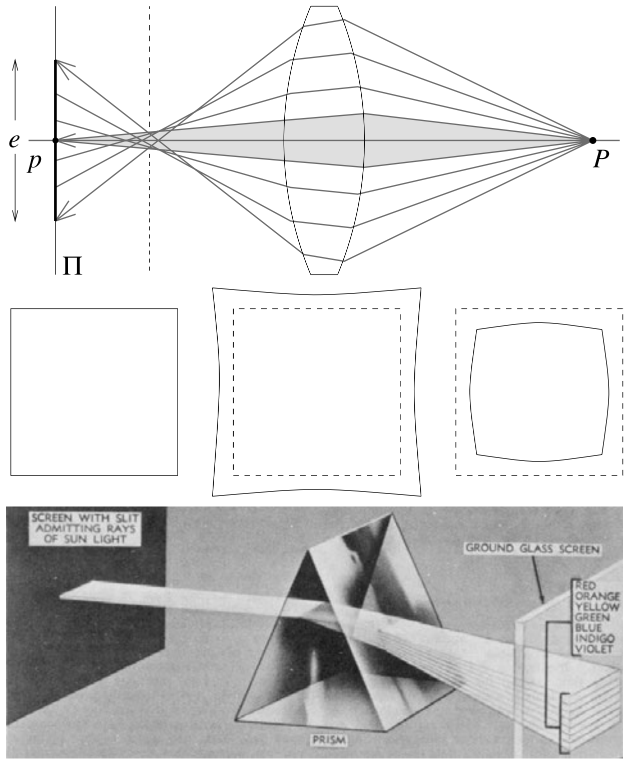

FIGURE 1.11: Aberrations. (a) Spherical aberration: The gray region is the paraxial zone where the rays issued from P intersect at its paraxial image p. If an image plane π were erected in p, the image of p in that plane would form a circle of confusion of diameter e. The focus plane yielding the circle of least confusion is indicated by a dashed line. (b) Distortion: From left to right, the nominal image of a fronto-parallel square, pincushion distortion, and barrel distortion. (c) Chromatic aberration: The index of refraction of a transparent medium depends on the wavelength (or color) of the incident light rays. Here, a prism decomposes white light into a palette of colors. *Figure from US NAVY MANUAL OF BASIC OPTICS AND OPTICAL INSTRUMENTS, prepared by the Bureau of Naval Personnel, reprinted by Dover Publications, Inc. (1969).*

Besides spherical aberration, there are four other types of *primary aberrations* caused by the differences between first- and third-order optics, namely *coma*, *astigmatism*, *field curvature*, and *distortion*. A precise definition of these aberrations is beyond the scope of this book. Suffice to say that, like a spherical aberration, the first three degrade the image by blurring the picture of every object point. Distortion, on the other hand, plays a different role and changes the shape of the image

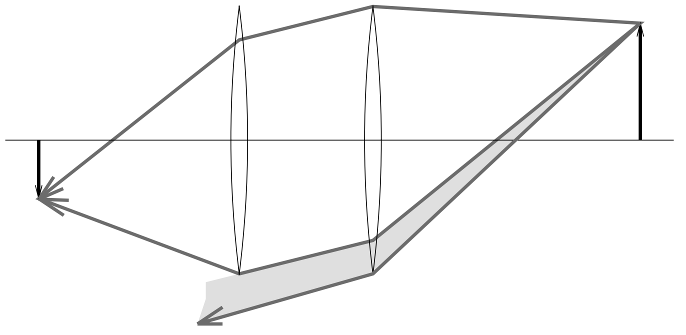

FIGURE 1.12: Vignetting effect in a two-lens system. The shaded part of the beam never reaches the second lens. Additional apertures and stops in a lens further contribute to vignetting.

as a whole (Figure 1.11 [b]). This effect is due to the fact that different areas of a lens have slightly different focal lengths. The aberrations mentioned so far are monochromatic, i.e., they are independent of the response of the lens to various wavelengths. However, the index of refraction of a transparent medium depends on wavelength (Figure 1.11 [c]), and it follows from the thin lens equation (Eq. [1.5]) that the focal length depends on wavelength as well. This causes the phenomenon of *chromatic aberration*: refracted rays corresponding to different wavelengths will intersect the optical axis at different points (longitudinal chromatic aberration) and form different circles of confusion in the same image plane (transverse chromatic aberration).

Aberrations can be minimized by aligning several simple lenses with well-chosen shapes and refraction indices, separated by appropriate stops. These *compound lenses* can still be modeled by the thick lens equations, but they suffer from one more defect relevant to machine vision: light beams emanating from object points located off-axis are partially blocked by the various apertures (including the individual lens components themselves) positioned inside the lens to limit aberrations (Figure 1.12). This phenomenon, called *vignetting*, causes the image brightness to drop in the image periphery. Vignetting may pose problems to automated image analysis programs, but it is not quite as important in photography, thanks to the human eye's remarkable insensitivity to smooth brightness gradients. Speaking of which, it is time to have a look at this extraordinary organ.

1.1.4 The Human Eye

Here we give a (brief) overview of the anatomical structure of the eye. It is largely based on the presentation in Wandell (1995), and the interested reader is invited to read this excellent book for more details. Figure 1.13 (left) is a sketch of the section of an eyeball through its vertical plane of symmetry, showing the main elements of the eye: the *iris* and the *pupil*, which control the amount of light penetrating the eyeball; the *cornea* and the crystalline *lens*, which together refract the light to create the retinal image; and finally the *retina*, where the image is formed. Despite its globular shape, the human eyeball is functionally similar to a camera with a field of view covering a 160° (width) × 135° (height) area. Like any other optical system, it suffers from various types of geometric and chromatic aberrations. Several models of the eye obeying the laws of first-order geometric optics have been proposed, and Figure 1.13 (right) shows one of them, *Helmoltz's schematic eye*. There are only three refractive surfaces, with an infinitely thin cornea and a homogeneous lens. The constants given in Figure 1.13 are for the eye focusing at infinity (*unaccommodated eye*). This model is of course only an approximation of the real optical characteristics of the eye.

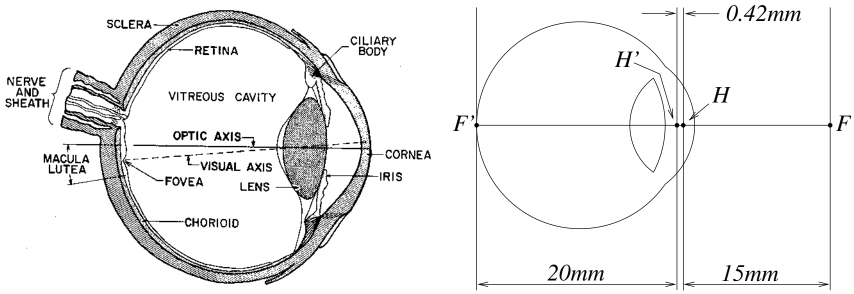

FIGURE 1.13: Left: the main components of the human eye. *Reproduced with permission, the American Society for Photogrammetry and Remote Sensing. A.L. Nowicki, "Stereoscopy." MANUAL OF PHOTOGRAMMETRY, edited by M.M. Thompson, R.C. Eller, W.A. Radlinski, and J.L. Speert, third edition, pp. 515–536. Bethesda: American Society of Photogrammetry, (1966).* Right: Helmoltz's schematic eye as modified by Laurance (after Driscoll and Vaughan, 1978). The distance between the pole of the cornea and the anterior principal plane is 1.96 mm, and the radii of the cornea, anterior, and posterior surfaces of the lens are respectively 8 mm, 10 mm, and 6 mm.

Let us have a second look at the components of the eye one layer at a time. The cornea is a transparent, highly curved, refractive window through which light enters the eye before being partially blocked by the colored and opaque surface of the iris. The pupil is an opening at the center of the iris whose diameter varies from about 1 to 8 mm in response to illumination changes, dilating in low light to increase the amount of energy that reaches the retina and contracting in normal lighting conditions to limit the amount of image blurring due to spherical aberration in the eye. The refracting power (reciprocal of the focal length) of the eye is, in large part, an effect of refraction at the the air–cornea interface, and it is fine-tuned by deformations of the crystalline lens that accommodates to bring objects into sharp focus. In healthy adults, it varies between 60 (unaccommodated case) and 68 diopters (1 diopter $= 1\ \mathrm{m}^{-1}$), corresponding to a range of focal lengths between 15 and 17 mm.

The retina itself is a thin, layered membrane populated by two types of photoreceptors—*rods* and *cones*. There are about 100 million rods and 5 million cones in a human eye. Their spatial distribution varies across the retina: The *macula lutea* is a region in the center of the retina where the concentration of cones is particularly high and images are sharply focused whenever the eye fixes its attention on an object (Figure 1.13). The highest concentration of cones occurs in the *fovea*, a depression in the middle of the macula lutea where it peaks at $1.6 \times 10^5/\text{mm}^2$, with the centers of two neighboring cones separated by only half a minute of visual angle. Conversely, there are no rods in the center of the fovea, but the rod density increases toward the periphery of the visual field. There is also a *blind spot* on the retina, where the ganglion cell axons exit the retina and form the optic nerve.

The rods are extremely sensitive photoreceptors, capable of responding to a single photon, but they yield relatively poor spatial detail despite their high number because many rods converge to the same neuron within the retina. In contrast, cones become active at higher light levels, but the signal output by each cone in the fovea is encoded by several neurons, yielding a high resolution in that area. As discussed further in Chapter 3, there are three types of cones with different spectral sensitivities, and these play a key role in the perception of color. Much more could (and should) be said about the human eye—for example how our two eyes verge and fixate on targets, and how they cooperate in stereo vision, an issue briefly discussed in Chapter 7.

1.2 INTRINSIC AND EXTRINSIC PARAMETERS

Digital images, like animal retinas, are spatially discrete, and divided into (usually) rectangular picture elements, or *pixels*. This is an aspect of the image formation process that we have neglected so far, assuming instead that the image domain is spatially continuous. Likewise, the perspective equation derived in the previous section is valid only when all distances are measured in the camera's reference frame, and when image coordinates have their origin at the image center where the axis of symmetry of the camera pierces its retina. In practice, the world and camera coordinate systems are related by a set of physical parameters, such as the focal length of the lens, the size of the pixels, the position of the image center, and the position and orientation of the camera. This section identifies these parameters. We will distinguish the *intrinsic* parameters, which relate the camera's coordinate system to the idealized coordinate system used in Section 1.1, from the *extrinsic* parameters, which relate the camera's coordinate system to a fixed world coordinate system and specify its position and orientation in space.

We ignore in the rest of this section the fact that, for cameras equipped with a lens, a point will be in focus only when its depth and the distance between the optical center of the camera and its image plane obey Eq. (1.5). In particular, we assume that the camera is focused at infinity, so $d = f$. Likewise, the nonlinear aberrations associated with real lenses are not taken into account by Eq. (1.1). We neglect these aberrations in this section, but revisit radial distortion in Section 1.3 when we address the problem of estimating the intrinsic and extrinsic parameters of a camera (a process known as *geometric camera calibration*).

1.2.1 Rigid Transformations and Homogeneous Coordinates

This section features our first use of *homogeneous* coordinates to represent the position of points in two or three dimensions. Consider a point P whose position in some coordinate frame $(F) = (O, \boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k})$ is given by

$$\overrightarrow{OP} = X\boldsymbol{i} + Y\boldsymbol{j} + Z\boldsymbol{k}.$$

We define the usual (nonhomogeneous) coordinate vector of P to be the vector $(X, Y, Z)^T$ in $\mathbb{R}^3$ and its homogeneous coordinate vector as the vector $(X, Y, Z, 1)^T$ in $\mathbb{R}^4$. We use bold letters to denote (homogeneous and nonhomogeneous) coordinate vectors in this book, and always state which type of coordinates we use when it is not obvious from the context. We also use a superscript *on the left side of* coordinate vectors when necessary to indicate which coordinate frame a position is expressed in. For example, ${}^F\boldsymbol{P}$ stands for the coordinate vector of the point P in the frame (F). Homogeneous coordinates are a convenient device for representing various geometric transformations by matrix products. For example, the change of coordinates between two Euclidean coordinate systems (A) and (B) may be represented by a 3×3 rotation matrix $\mathcal{R}$ and a translation vector $\boldsymbol{t}$ in $\mathbb{R}^3$, and the corresponding *rigid transformation* can be written in nonhomogeneous coordinates as

$$ {}^A\boldsymbol{P} = \mathcal{R}\,{}^B\boldsymbol{P} + \boldsymbol{t}, \tag{1.6} $$

where ${}^A\boldsymbol{P}$ and ${}^B\boldsymbol{P}$ are elements of $\mathbb{R}^3$. In homogeneous coordinates, we write instead

$$ {}^A\boldsymbol{P} = \mathcal{T}\,{}^B\boldsymbol{P}, \quad \text{where} \quad \mathcal{T} = \begin{pmatrix} \mathcal{R} & \boldsymbol{t} \\ \boldsymbol{0}^T & 1 \end{pmatrix}, \tag{1.7} $$

and ${}^A\boldsymbol{P}$ and ${}^B\boldsymbol{P}$ are this time elements of $\mathbb{R}^4$.

Before going further, let us recall a few facts about rotations. Rotation matrices form a mulitplicative group. From an analytical viewpoint, they are characterized by the facts that (1) the inverse of a rotation matrix is equal to its transpose, and (2) its determinant is equal to one. It can also be shown that any rotation matrix can be parameterized by three *Euler angles*, or written as the product of three elementary rotations about the $\boldsymbol{i}$, $\boldsymbol{j}$, and $\boldsymbol{k}$ vectors of some coordinate system. As shown in Chapters 7 and 14, other parameterizations—by exponentials of antisymmetric matrices or quaternions for example—may prove useful as well. Geometrically, the matrix $\mathcal{R}$ in Eq. (1.6) also represents the basis vectors $(\boldsymbol{i}_B, \boldsymbol{j}_B, \boldsymbol{k}_B)$ of (B) in the coordinate frame (A)—that is, the matrix $\mathcal{R}$ in Eq. (1.6) is given by:

$$ \mathcal{R} \stackrel{\text{def}}{=} \left({}^A\boldsymbol{i}_B, {}^A\boldsymbol{j}_B, {}^A\boldsymbol{k}_B\right) = \begin{pmatrix} \boldsymbol{i}_A \cdot \boldsymbol{i}_B & \boldsymbol{j}_A \cdot \boldsymbol{i}_B & \boldsymbol{k}_A \cdot \boldsymbol{i}_B \\ \boldsymbol{i}_A \cdot \boldsymbol{j}_B & \boldsymbol{j}_A \cdot \boldsymbol{j}_B & \boldsymbol{k}_A \cdot \boldsymbol{j}_B \\ \boldsymbol{i}_A \cdot \boldsymbol{k}_B & \boldsymbol{j}_A \cdot \boldsymbol{k}_B & \boldsymbol{k}_A \cdot \boldsymbol{k}_B \end{pmatrix}, \tag{1.8} $$

and, as shown in the problems at the end of this chapter, Eq. (1.6) easily follows from this definition. By definition, the columns of a rotation matrix form a right-handed orthonormal coordinate system of $\mathbb{R}^3$. It follows from properties (1) and (2) that their rows also form such a coordinate system. One may wonder what happens when $\mathcal{R}$ is replaced in Eq. (1.7) by some arbitrary nonsingular 3×3 matrix, or when the matrix $\mathcal{T}$ itself is replaced by some arbitrary nonsingular 4×4 matrix. As further discussed in Chapter 8, the coordinate frames (A) and (B) are no longer separated by rigid transformations in this case, but by *affine* and *projective transformations* respectively.

As will be shown in the rest of this section, homogeneous coordinates also provide an algrebraic representation of the perspective projection process in the form of a 3×4 matrix $\mathcal{M}$, so that the coordinate vector $\boldsymbol{P} = (X, Y, Z, 1)^T$ of a point P in some fixed world coordinate system and the coordinate vector $\boldsymbol{p} = (x, y, 1)^T$ of

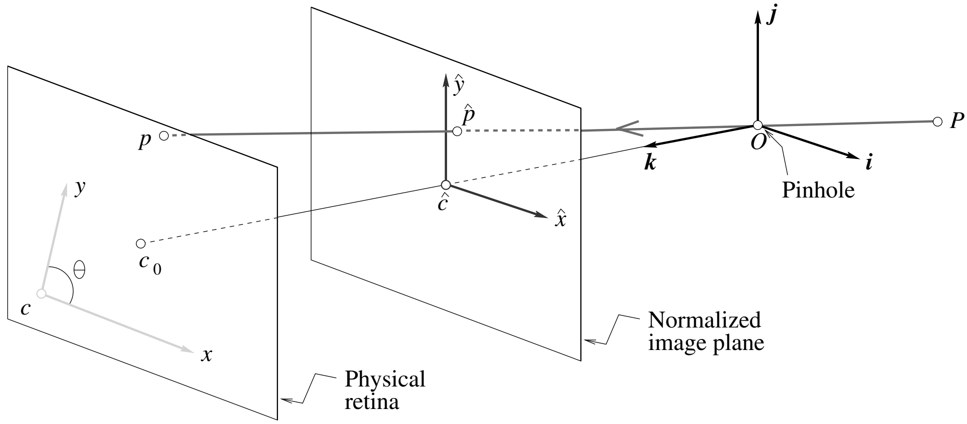

FIGURE 1.14: Physical and normalized image coordinate systems.

its image p in the camera's reference frame are related by the *perspective projection equation*

$$\boldsymbol{p} = \frac{1}{Z}\mathcal{M}\boldsymbol{P}. \tag{1.9}$$

1.2.2 Intrinsic Parameters

It is possible to associate with a camera a *normalized image plane* parallel to its physical retina but located at a unit distance from the pinhole. We attach to this plane its own coordinate system with an origin located at the point $\hat{c}$ where the optical axis pierces it (Figure 1.14). Equation (1.1) can be written in this normalized coordinate system as

$$\begin{cases} \hat{x} = \dfrac{X}{Z} \\ \hat{y} = \dfrac{Y}{Z} \end{cases} \iff \hat{\boldsymbol{p}} = \frac{1}{Z}\begin{pmatrix}\mathrm{Id} & \mathbf{0}\end{pmatrix}\boldsymbol{P}, \tag{1.10}$$

where $\hat{\boldsymbol{p}} \stackrel{\text{def}}{=} (\hat{x}, \hat{y}, 1)^T$ is the vector of homogeneous coordinates of the projection $\hat{p}$ of the point P into the normalized image plane, and $\boldsymbol{P}$ is as before the homogeneous coordinate vector of P in the world coordinate frame.

The physical retina of the camera is in general different (Figure 1.14): It is located at a distance $f \neq 1$ from the pinhole (remember that we assume that the camera is focused at infinity, so the distance between the pinhole and the image plane is equal to the focal length), and the coordinates (x, y) of the image point p are usually expressed in pixel units (instead of, say, meters). In addition, pixels may be rectangular instead of square, so the camera has two additional scale parameters

k and l, and

$$\begin{cases} x = kf\dfrac{X}{Z} = kf\hat{x}, \\ y = lf\dfrac{Y}{Z} = lf\hat{y}. \end{cases} \tag{1.11}$$

Let us talk units for a second: f is a distance, expressed in meters, for example, and a pixel will have dimensions $\frac{1}{k} \times \frac{1}{l}$, where k and l are expressed in pixel $\times\, \mathrm{m}^{-1}$. The parameters k, l, and f are not independent, and they can be replaced by the magnifications $\alpha = kf$ and $\beta = lf$ expressed in pixel units.

Now, in general, the actual origin of the camera coordinate system is at a corner c of the retina (in the case depicted in Figure 1.14, the lower-left corner, or sometimes the upper-left corner, when the image coordinates are the row and column indices of a pixel) and not at its center, and the center of the CCD matrix usually does not coincide with the image center c_0. This adds two parameters x_0 and y_0 that define the position (in pixel units) of c_0 in the retinal coordinate system. Thus, Eq. (1.11) is replaced by

$$\begin{cases} x = \alpha\hat{x} + x_0, \\ y = \beta\hat{y} + y_0. \end{cases} \tag{1.12}$$

Finally, the camera coordinate system might also be skewed, due to some manufacturing error, so the angle θ between the two image axes is not equal to (but of course not very different from) 90 degrees. In this case, it is easy to show that Eq. (1.12) transforms into

$$\begin{cases} x = \alpha\hat{x} - \alpha\cot\theta\hat{y} + x_0, \\ y = \dfrac{\beta}{\sin\theta}\hat{y} + y_0. \end{cases} \tag{1.13}$$

This can be written in matrix form as

$$\boldsymbol{p} = \mathcal{K}\hat{\boldsymbol{p}}, \quad \text{where} \quad \boldsymbol{p} = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \quad \text{and} \quad \mathcal{K} \stackrel{\text{def}}{=} \begin{pmatrix} \alpha & -\alpha\cot\theta & x_0 \\ 0 & \dfrac{\beta}{\sin\theta} & y_0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{1.14}$$

The 3×3 matrix $\mathcal{K}$ is called the (internal) *calibration matrix* of the camera. Putting Eqs. (1.10) and (1.14) together, we obtain

$$\boldsymbol{p} = \frac{1}{Z}\mathcal{K}\big(\mathrm{Id} \quad \mathbf{0}\big)\boldsymbol{P} = \frac{1}{Z}\mathcal{M}\boldsymbol{P}, \quad \text{where} \quad \mathcal{M} \stackrel{\text{def}}{=} \big(\mathcal{K} \quad \mathbf{0}\big), \tag{1.15}$$

which is indeed an instance of Eq. (1.9). The five parameters α, β, θ, x_0, and y_0 are called the *intrinsic parameters* of the camera.

Several of these parameters, such as the focal length, or the physical size of the pixels, are often available in the *EXIF tags* attached to the JPEG images recorded by digital cameras (this information might not be available, of course, as in the case of stock film footage). For zoom lenses, the focal length may vary with time, along with the image center when the optical axis of the lens is not exactly perpendicular to the image plane. Simply changing the focus of the camera will also affect the magnification because it will change the lens-to-retina distance, but we will continue to assume that the camera is focused at infinity and ignore this effect in the rest of this chapter.

1.2.3 Extrinsic Parameters

Equation (1.15) is written in a coordinate frame (C) attached to the camera. Let us now consider the case where this frame is distinct from the world coordinate system (W). To emphasize this, we rewrite Eq. (1.15) as $\boldsymbol{p} = \frac{1}{Z}\mathcal{M}\,{}^C\boldsymbol{P}$, where ${}^C\boldsymbol{P}$ denotes the vector of homogeneous coordinates of the point P expressed in (C). The change of coordinates between (C) and (W) is a rigid transformation, and it can be written as

$${}^C\boldsymbol{P} = \begin{pmatrix} \mathcal{R} & \boldsymbol{t} \\ & \\ \boldsymbol{0}^T & 1 \end{pmatrix} {}^W\boldsymbol{P},$$

where ${}^W\boldsymbol{P}$ is the vector of homogeneous coordinates of the point P in the coordinate frame (W). Taking $\boldsymbol{P} = {}^W\boldsymbol{P}$ and substituting in Eq. (1.15) finally yields

$$\boldsymbol{p} = \frac{1}{Z}\mathcal{M}\boldsymbol{P}, \quad \text{where} \quad \mathcal{M} = \mathcal{K}\begin{pmatrix}\mathcal{R} & \boldsymbol{t}\end{pmatrix}. \tag{1.16}$$

This is the most general form of the perspective projection equation, and indeed an instance of Eq. (1.9). Knowing $\mathcal{M}$ determines the position of the camera's optical center in the coordinate frame (W)—that is, its homogeneous coordinate vector $\boldsymbol{O} = {}^W\boldsymbol{O}$. Indeed, as shown in the problems at the end of this chapter, $\mathcal{M}\boldsymbol{O} = \boldsymbol{0}$.

As mentioned earlier, a rotation matrix such as $\mathcal{R}$ is defined by three independent parameters (for example, Euler angles). Adding to these the three coordinates of the vector $\boldsymbol{t}$, we obtain a set of six *extrinsic parameters* that define the position and orientation of the camera relative to the world coordinate frame.

It is very important to understand that the depth Z in Eq. (1.16) is *not* independent of $\mathcal{M}$ and $\boldsymbol{P}$, because if $\boldsymbol{m}_1^T$, $\boldsymbol{m}_2^T$ and $\boldsymbol{m}_3^T$ denote the three rows of $\mathcal{M}$, it follows directly from Eq. (1.16) that $Z = \boldsymbol{m}_3 \cdot \boldsymbol{P}$. In fact, it is sometimes convenient to rewrite Eq. (1.16) in the equivalent form:

$$\begin{cases} x = \dfrac{\boldsymbol{m}_1 \cdot \boldsymbol{P}}{\boldsymbol{m}_3 \cdot \boldsymbol{P}}, \\[2ex] y = \dfrac{\boldsymbol{m}_2 \cdot \boldsymbol{P}}{\boldsymbol{m}_3 \cdot \boldsymbol{P}}. \end{cases} \tag{1.17}$$

A perspective projection matrix can be written explicitly as a function of its five intrinsic parameters, the three rows $\boldsymbol{r}_1^T$, $\boldsymbol{r}_2^T$, and $\boldsymbol{r}_3^T$ of the matrix $\mathcal{R}$, and the three coordinates t_1, t_2, and t_3 of the vector $\boldsymbol{t}$, namely:

$$\mathcal{M} = \begin{pmatrix} \alpha\boldsymbol{r}_1^T - \alpha\cot\theta\boldsymbol{r}_2^T + x_0\boldsymbol{r}_3^T & \alpha t_1 - \alpha\cot\theta t_2 + x_0 t_3 \\ \dfrac{\beta}{\sin\theta}\boldsymbol{r}_2^T + y_0\boldsymbol{r}_3^T & \dfrac{\beta}{\sin\theta}t_2 + y_0 t_3 \\ \boldsymbol{r}_3^T & t_3 \end{pmatrix}. \tag{1.18}$$

When $\mathcal{R}$ is written as the product of three elementary rotations, the vectors $\boldsymbol{r}_i$ $(i = 1, 2, 3)$ can of course be written in terms of the corresponding three angles, and Eq. (1.18) gives an explicit parameterization of $\mathcal{M}$ in terms of all 11 camera parameters.

1.2.4 Perspective Projection Matrices

This section examines the conditions under which a 3×4 matrix $\mathcal{M}$ can be written in the form given by Eq. (1.18). Let us write without loss of generality $\mathcal{M} = \begin{pmatrix}\mathcal{A} & \boldsymbol{b}\end{pmatrix}$, where $\mathcal{A}$ is a 3×3 matrix and $\boldsymbol{b}$ is an element of $\mathbb{R}^3$, and let us denote by $\boldsymbol{a}_3^T$ the third row of $\mathcal{A}$. Clearly, if $\mathcal{M}$ is an instance of Eq. (1.18), then $\boldsymbol{a}_3^T$ must be a unit vector since it is equal to $\boldsymbol{r}_3^T$, the last row of a rotation matrix. Note, however, that replacing $\mathcal{M}$ by $\lambda\mathcal{M}$ in Eq. (1.17) for some arbitrary $\lambda \neq 0$ does not change the corresponding image coordinates. This will lead us in the rest of this book to consider projection matrices as *homogeneous objects*, only defined up to scale, whose canonical form, as expressed by Eq. (1.18), can be obtained by choosing a scale factor such that $||\boldsymbol{a}_3|| = 1$. Note that the parameter Z in Eq. (1.16) can only rightly be interpreted as the depth of the point P when $\mathcal{M}$ is written in this canonical form. Note also that the number of intrinsic and extrinsic parameters of a camera matches the 11 free parameters of the (homogeneous) matrix $\mathcal{M}$.

We say that a 3×4 matrix that can be written (up to scale) as Eq. (1.18) for some set of intrinsic and extrinsic parameters is a *perspective projection matrix*. It is of practical interest to put some restrictions on the intrinsic parameters of a camera because, as noted earlier, some of these parameters will be fixed and might be known. In particular, we will say that a 3×4 matrix is a *zero-skew perspective projection matrix* when it can be rewritten (up to scale) as Eq. (1.18) with $\theta = \pi/2$, and that it is a *perspective projection matrix with zero skew and unit aspect-ratio* when it can be rewritten (up to scale) as Eq. (1.18) with $\theta = \pi/2$ and $\alpha = \beta$. A camera with *known* nonzero skew and nonunit aspect-ratio can be transformed into a camera with zero skew and unit aspect-ratio by an appropriate change of image coordinates. Are arbitrary 3×4 matrices perspective projection matrices? The following theorem answers this question.

Theorem 1. Let $\mathcal{M} = \begin{pmatrix}\mathcal{A} & \boldsymbol{b}\end{pmatrix}$ be a 3×4 matrix, and let $\boldsymbol{a}_i^T$ $(i = 1, 2, 3)$ denote the rows of the matrix $\mathcal{A}$ formed by the three leftmost columns of $\mathcal{M}$.

- A necessary and sufficient condition for $\mathcal{M}$ to be a perspective projection matrix is that $\mathrm{Det}(\mathcal{A}) \neq 0$.
- A necessary and sufficient condition for $\mathcal{M}$ to be a zero-skew perspective projection matrix is that $\mathrm{Det}(\mathcal{A}) \neq 0$ and

$$(\boldsymbol{a}_1 \times \boldsymbol{a}_3) \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3) = 0.$$

- A necessary and sufficient condition for $\mathcal{M}$ to be a perspective projection matrix with zero skew and unit aspect-ratio is that $\mathrm{Det}(\mathcal{A}) \neq 0$ and

$$\begin{cases} (\boldsymbol{a}_1 \times \boldsymbol{a}_3) \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3) = 0, \\ (\boldsymbol{a}_1 \times \boldsymbol{a}_3) \cdot (\boldsymbol{a}_1 \times \boldsymbol{a}_3) = (\boldsymbol{a}_2 \times \boldsymbol{a}_3) \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3). \end{cases}$$

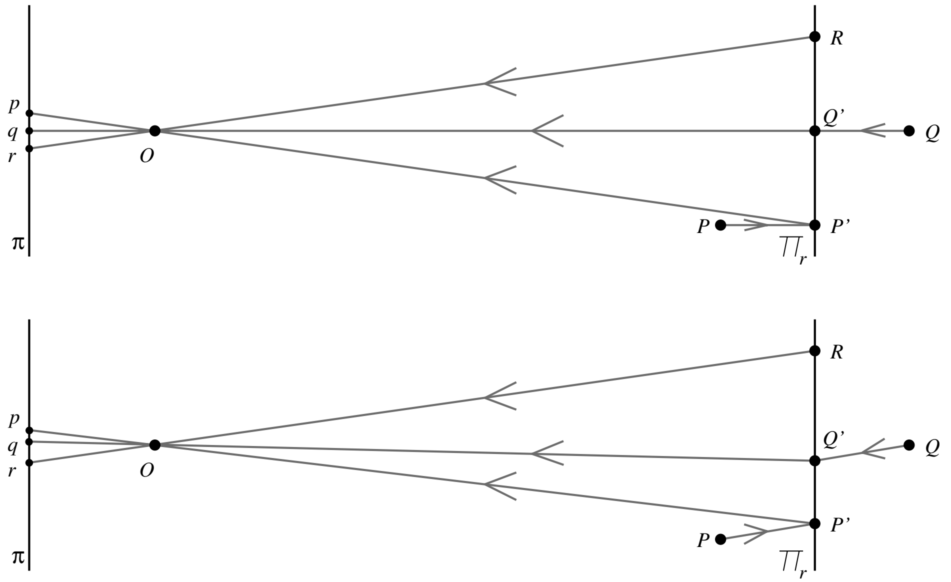

FIGURE 1.15: Affine projection models: (top) weak-perspective and (bottom) paraperspective projections.

The conditions of the theorem are clearly necessary: By definition, given some perspective projection matrix $\mathcal{A}$, we can always write $\rho\mathcal{A} = \mathcal{K}\mathcal{R}$ for some nonzero scalar ρ, calibration matrix $\mathcal{K}$, rotation matrix $\mathcal{R}$, and vector $\boldsymbol{t}$. In particular, $\rho^3 \mathrm{Det}(\mathcal{A}) = \mathrm{Det}(\mathcal{K}) \neq 0$ since calibration matrices are nonsingular by construction, so $\mathcal{A}$ is nonsingular. Further, a simple calculation shows that the rows of the matrix $\frac{1}{\rho}\mathcal{K}\mathcal{R}$ satisfy the conditions of the theorem under the various assumptions imposed by its statement. These conditions are proven to also be sufficient in Faugeras (1993).

1.2.5 Weak-Perspective Projection Matrices

As noted in Section 1.1.2, when a scene's relief is small compared to the overall distance separating it from the camera observing it, a weak-perspective projection model can be used to approximate the imaging process (Figure 1.15, top). Let O denote the optical center of the camera, and let R denote a scene reference point. The weak-perspective projection of a scene point P is constructed in two steps: the point P is first projected orthogonally onto a point P' of the plane Π_r parallel to the image plane Π and passing through R; perspective projection is then used to map the point P' onto the image point p. Since π_r is a fronto-parallel plane, the net effect of the second projection step is a scaling of the image coordinates.

As shown in this section, the weak-perspective projection process can be represented in terms of a 2×4 matrix $\mathcal{M}$, so that the *homogeneous* coordinate vector $\boldsymbol{P} = (X, Y, Z, 1)^T$ of a point P in some fixed world coordinate system and the *non-homogeneous* coordinate vector $\boldsymbol{p} = (x, y)^T$ of its image p in the camera's reference frame are related by the *affine projection equation*

$$\boldsymbol{p} = \mathcal{M}\boldsymbol{P}. \tag{1.19}$$

It turns out that this general model accomodates various other approximations of the perspective projection process. These include the orthographic projection model discussed earlier, as well as the *parallel projection* model, which subsumes the orthographic one, and takes into account the fact that the objects of interest may lie off the optical axis of the camera. In this model, the viewing rays are parallel to each other but not necessarily perpendicular to the image plane. Paraperspective is another affine projection model that takes into account both the distortions associated with a reference point that is off the optical axis of the camera and possible variations in depth (Figure 1.15, bottom). Using the same notation as before, and denoting by Δ the line joining the optical center O to the reference point R, parallel projection in the direction of Δ is first used to map P onto a point P' of the plane Π_r; perspective projection is then used to map the point P' onto the image point p.

We will focus on weak perspective in the rest of this section. Let us derive the corresponding projection equation. If Z_r denotes the depth of the reference point R, the two elementary projection stages $P \to P' \to p$ can be written in the normalized coordinate system attached to the camera as

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \longrightarrow \begin{pmatrix} Z \\ Y \\ Z_r \end{pmatrix} \longrightarrow \begin{pmatrix} \hat{x} \\ \hat{y} \\ 1 \end{pmatrix} = \begin{pmatrix} X/Z_r \\ Y/Z_r \\ 1 \end{pmatrix},$$

or, in matrix form,

$$\begin{pmatrix} \hat{x} \\ \hat{y} \\ 1 \end{pmatrix} = \frac{1}{Z_r} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & Z_r \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}.$$

Introducing the calibration matrix $\mathcal{K}$ of the camera and its extrinsic parameters $\mathcal{R}$ and $\boldsymbol{t}$ gives the general form of the projection equation, i.e.,

$$\boldsymbol{p} = \frac{1}{Z_r}\mathcal{K} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & Z_r \end{pmatrix} \begin{pmatrix} \mathcal{R} & \boldsymbol{t} \\ \boldsymbol{0}^T & 1 \end{pmatrix} \boldsymbol{P}, \tag{1.20}$$

where $\boldsymbol{P}$ and $\boldsymbol{p}$ denote as before the homogeneous coordinate vector of the point P in the world reference frame, and the homogeneous coordinate vector of its projection p in the camera's coordinate system. Finally, noting that Z_r is a constant and writing

$$\mathcal{K} = \begin{pmatrix} \mathcal{K}_2 & \boldsymbol{p}_0 \\ \boldsymbol{0}^T & 1 \end{pmatrix}, \quad \text{where} \quad \mathcal{K}_2 \stackrel{\text{def}}{=} \begin{pmatrix} \alpha & -\alpha \cot\theta \\ 0 & \dfrac{\beta}{\sin\theta} \end{pmatrix} \quad \text{and} \quad \boldsymbol{p}_0 \stackrel{\text{def}}{=} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix},$$

allows us to rewrite Eq. (1.20) as

$$\boldsymbol{p} = \mathcal{M}\boldsymbol{P}, \quad \text{where} \quad \mathcal{M} = \begin{pmatrix} \mathcal{A} & \boldsymbol{b} \end{pmatrix}, \tag{1.21}$$

where $\boldsymbol{p}$ is, this time, the *nonhomogeneous* coordinate vector of the point p, and $\mathcal{M}$ is a 2×4 projection matrix (compare to the general perspective case of Eq. [1.16]). In this expression, the 2×3 matrix $\mathcal{A}$ and the 2-vector $\boldsymbol{b}$ are respectively defined by

$$\mathcal{A} = \frac{1}{Z_r}\mathcal{K}_2\mathcal{R}_2 \quad \text{and} \quad \boldsymbol{b} = \frac{1}{Z_r}\mathcal{K}_2\boldsymbol{t}_2 + \boldsymbol{p}_0,$$

where $\mathcal{R}_2$ denotes the 2×3 matrix formed by the first two rows of $\mathcal{R}$, and $\boldsymbol{t}_2$ denotes the 2-vector formed by the first two coordinates of $\boldsymbol{t}$.

Note that t_3 does not appear in the expression of $\mathcal{M}$, and that $\boldsymbol{t}_2$ and $\boldsymbol{p}_0$ are coupled in this expression: the projection matrix does not change when $\boldsymbol{t}_2$ is replaced by $\boldsymbol{t}_2 + \boldsymbol{a}$ and $\boldsymbol{p}_0$ is replaced by $\boldsymbol{p}_0 - \frac{1}{Z_r}\mathcal{K}_2\boldsymbol{a}$. This redundancy allows us to arbitrarily choose $x_0 = y_0 = 0$. In other words, the position of the center of the image is immaterial for weak-perspective projection. Note that the values of Z_r, α, and β are also coupled in the expression of $\mathcal{M}$, and that the value of Z_r is a priori unknown in most applications. This allows us to write

$$\mathcal{M} = \frac{1}{Z_r}\begin{pmatrix} k & s \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \mathcal{R}_2 & \boldsymbol{t}_2 \end{pmatrix}, \tag{1.22}$$

where k and s denote the aspect ratio and the skew of the camera, respectively. In particular, a weak-perspective projection matrix is defined by two intrinsic parameters (k and s), five extrinsic parameters (the three angles defining $\mathcal{R}_2$ and the two coordinates of $\boldsymbol{t}_2$), and one scene-dependent *structure* parameter Z_r.

A 2×4 matrix $\mathcal{M} = \begin{pmatrix} \mathcal{A} & \boldsymbol{b} \end{pmatrix}$ where $\mathcal{A}$ is an arbitrary rank-2 2×3 matrix and $\boldsymbol{b}$ is an arbitrary vector in $\mathbb{R}^2$ is called an *affine projection matrix*. Both weak-perspective and general affine projection matrices are defined by eight independent parameters. Weak-perspective projection matrices are affine ones of course. Conversely, a simple parameter-counting argument suggests that it should be possible to write an arbitrary affine projection matrix as a weak-perspective one. This is confirmed by the following theorem.

Theorem 2. An affine projection matrix can be written uniquely (up to a sign ambiguity) as a general weak-perspective projection matrix as defined by Eq. (1.22).

This theorem is proven in Faugeras *et al.* (2001, Propositions 4.26 and 4.27) and the problems.

1.3 GEOMETRIC CAMERA CALIBRATION

This section addresses the problem of estimating the intrinsic and extrinsic parameters of a camera from the image positions of scene features such as points of lines, whose positions are known in some fixed world coordinate system (Figure 1.16). In this context, camera calibration can be modeled as an optimization process, where the discrepancy between the observed image features and their theoretical positions is minimized with respect to the camera's intrinsic and extrinsic parameters.

Specifically, we assume that the image positions (x_i, y_i) of n fiducial points P_i $(i = 1, \ldots, n)$ with known homogeneous coordinate vectors $\boldsymbol{P}_i$ have been found

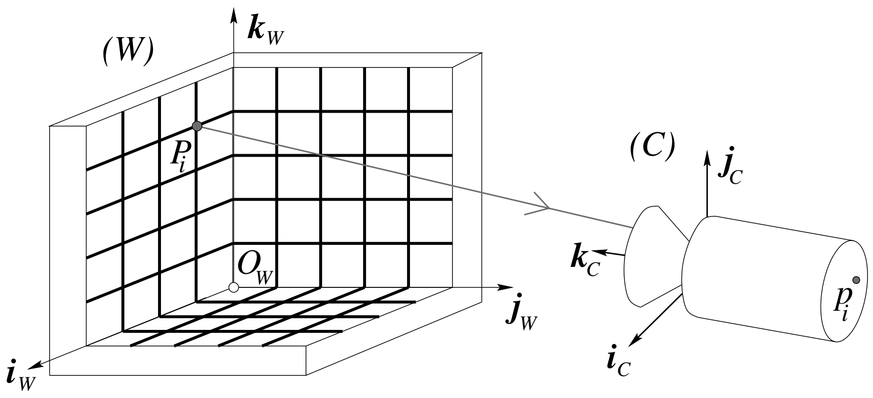

FIGURE 1.16: Camera calibration setup: In this example, the calibration rig is formed by three grids drawn in orthogonal planes. Other patterns could be used as well, and they may involve lines or other geometric figures.

in a picture of a calibration rig, either automatically or by hand. In the absence of modeling and measurement errors, geometric camera calibration amounts to finding the intrinsic and extrinsic parameters $\boldsymbol{\xi}$ such that

$$\begin{cases} x_i = \dfrac{\boldsymbol{m}_1(\boldsymbol{\xi}) \cdot \boldsymbol{P}_i}{\boldsymbol{m}_3(\boldsymbol{\xi}) \cdot \boldsymbol{P}_i}, \\ y_i = \dfrac{\boldsymbol{m}_2(\boldsymbol{\xi}) \cdot \boldsymbol{P}_i}{\boldsymbol{m}_3(\boldsymbol{\xi}) \cdot \boldsymbol{P}_i}, \end{cases} \tag{1.23}$$

where $\boldsymbol{m}_i^T(\boldsymbol{\xi})$ denotes the i^{th} row of the projection matrix $\mathcal{M}$, explicitly parameterized in this equation by the camera parameters. In the typical case where there are more measurements than unknowns (at least six points for 11 intrinsic and extrinsic parameters), Eq. (1.23) does not admit an exact solution, and an approximate one has to be found as the solution of a *least-squares* minimization problem (see Chapter 22). We present two least-squares formulations of the calibration problem in the rest of this section. The corresponding algorithms are illustrated with the calibration data shown in Figure 1.17.

1.3.1 A Linear Approach to Camera Calibration

We decompose the calibration process into (1) the computation of the perspective projection matrix $\mathcal{M}$ associated with the camera, followed by (2) the estimation of the intrinsic and extrinsic parameters of the camera from this matrix.

Estimation of the Projection Matrix. Let us assume that our camera has nonzero skew. According to Theorem 1, the matrix $\mathcal{M}$ is not singular, but otherwise arbitrary. Clearing the denominators in Eq. (1.23) yields two *linear* equations in $\boldsymbol{m}_1$, $\boldsymbol{m}_2$, and $\boldsymbol{m}_3$ (we omit the parameters $\boldsymbol{\xi}$ from now on for the sake of con-

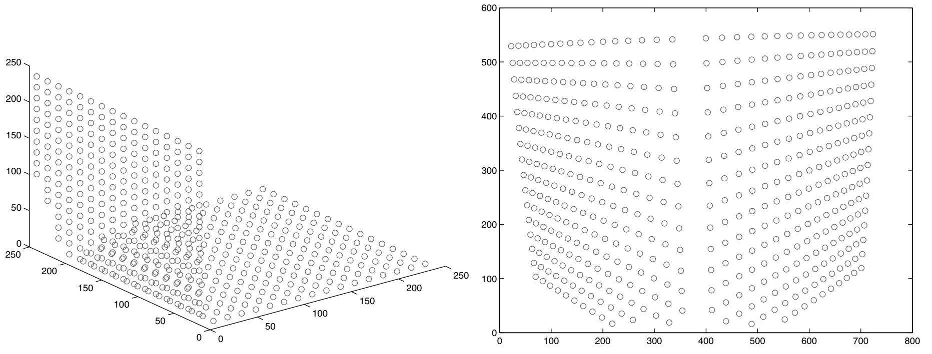

FIGURE 1.17: Camera calibration data. Left: A rendering of 491 3D fiducial points measured on a calibration rig. Right: The corresponding image points. Data courtesy of Janne Heikkilä; data copyright ©2000 University of Oulu.

ciseness), namely

$$\begin{cases} (\boldsymbol{m}_1 - x_i \boldsymbol{m}_3) \cdot \boldsymbol{P}_i & = & \boldsymbol{P}_i^T \boldsymbol{m}_1 + \mathbf{0}^T \boldsymbol{m}_2 - x_i \boldsymbol{P}_i^T \boldsymbol{m}_3 & = & 0, \\ (\boldsymbol{m}_2 - y_i \boldsymbol{m}_3) \cdot \boldsymbol{P}_i & = & \mathbf{0}^T \boldsymbol{m}_1 + \boldsymbol{P}_i^T \boldsymbol{m}_2 - y_i \boldsymbol{P}_i^T \boldsymbol{m}_3 & = & 0. \end{cases}$$

Collecting the constraints associated with all points yields a system of $2n$ homogeneous linear equations in the 12 coefficients of the matrix $\mathcal{M}$, namely,

$$\mathcal{P}\boldsymbol{m} = 0, \tag{1.24}$$

where

$$\mathcal{P} \stackrel{\text{def}}{=} \begin{pmatrix} \boldsymbol{P}_1^T & \mathbf{0}^T & -x_1 \boldsymbol{P}_1^T \\ \mathbf{0}^T & \boldsymbol{P}_1^T & -y_1 \boldsymbol{P}_1^T \\ \dots & \dots & \dots \\ \boldsymbol{P}_n^T & \mathbf{0}^T & -x_n \boldsymbol{P}_n^T \\ \mathbf{0}^T & \boldsymbol{P}_n^T & -y_n \boldsymbol{P}_n^T \end{pmatrix} \quad \text{and} \quad \boldsymbol{m} \stackrel{\text{def}}{=} \begin{pmatrix} \boldsymbol{m}_1 \\ \boldsymbol{m}_2 \\ \boldsymbol{m}_3 \end{pmatrix} = 0.$$

When $n \geq 6$, homogeneous linear least-squares can be used to compute the value of the unit vector $\boldsymbol{m}$ (hence the matrix $\mathcal{M}$) that minimizes $||\mathcal{P}\boldsymbol{m}||^2$ as the eigenvector of the 12×12 matrix $\mathcal{P}^T\mathcal{P}$ associated with its smallest eigenvalue (see Chapter 22). Note that any nonzero multiple of the vector $\boldsymbol{m}$ would have done just as well, reflecting the fact that $\mathcal{M}$ is defined by only 11 independent parameters.

Degenerate Point Configurations. Before showing how to recover the intrinsic and extrinsic parameters of the camera, let us pause to examine the *degenerate configurations* of the points P_i ($i = 1, \dots, n$) that may cause the failure of the camera calibration process. We focus on the (ideal) case where the positions $\boldsymbol{p}_i$ ($i = 1, \dots, n$) of the image points can be measured with zero error, and identify the *nullspace* of the matrix $\mathcal{P}$ (i.e., the subspace of $\mathbb{R}^{12}$ formed by the vectors $\boldsymbol{l}$ such that $\mathcal{P}\boldsymbol{l} = \mathbf{0}$).

Let $\boldsymbol{l}$ be such a vector. Introducing the vectors formed by successive quadruples of its coordinates—that is, $\boldsymbol{\lambda} = (l_1, l_2, l_3, l_4)^T$, $\boldsymbol{\mu} = (l_5, l_6, l_7, l_8)^T$, and

$\boldsymbol{\nu} = (l_9, l_{10}, l_{11}, l_{12})^T$—allows us to write

$$\mathbf{0} = \mathcal{P}\boldsymbol{l} = \begin{pmatrix} \boldsymbol{P}_1^T & \mathbf{0}^T & -x_1\boldsymbol{P}_1^T \\ \mathbf{0}^T & \boldsymbol{P}_1^T & -y_1\boldsymbol{P}_1^T \\ \dots & \dots & \dots \\ \boldsymbol{P}_n^T & \mathbf{0}^T & -x_n\boldsymbol{P}_n^T \\ \mathbf{0}^T & \boldsymbol{P}_n^T & -y_n\boldsymbol{P}_n^T \end{pmatrix} \begin{pmatrix} \boldsymbol{\lambda} \\ \boldsymbol{\mu} \\ \boldsymbol{\nu} \end{pmatrix} = \begin{pmatrix} \boldsymbol{P}_1^T\boldsymbol{\lambda} - x_1\boldsymbol{P}_1^T\boldsymbol{\nu} \\ \boldsymbol{P}_1^T\boldsymbol{\mu} - y_1\boldsymbol{P}_1^T\boldsymbol{\nu} \\ \dots \\ \boldsymbol{P}_n^T\boldsymbol{\lambda} - x_n\boldsymbol{P}_n^T\boldsymbol{\nu} \\ \boldsymbol{P}_n^T\boldsymbol{\mu} - y_n\boldsymbol{P}_n^T\boldsymbol{\nu} \end{pmatrix}. \tag{1.25}$$

Combining Eq. (1.23) with Eq. (1.25) yields

$$\begin{cases} \boldsymbol{P}_i^T\boldsymbol{\lambda} - \dfrac{\boldsymbol{m}_1^T\boldsymbol{P}_i}{\boldsymbol{m}_3^T\boldsymbol{P}_i}\boldsymbol{P}_i^T\boldsymbol{\nu} = 0, \\ \boldsymbol{P}_i^T\boldsymbol{\mu} - \dfrac{\boldsymbol{m}_2^T\boldsymbol{P}_i}{\boldsymbol{m}_3^T\boldsymbol{P}_i}\boldsymbol{P}_i^T\boldsymbol{\nu} = 0, \end{cases} \quad \text{for} \quad i = 1, \dots, n.$$

Thus, after clearing the denominators and rearranging the terms, we finally obtain:

$$\begin{cases} \boldsymbol{P}_i^T(\boldsymbol{\lambda}\boldsymbol{m}_3^T - \boldsymbol{m}_1\boldsymbol{\nu}^T)\boldsymbol{P}_i = 0, \\ \boldsymbol{P}_i^T(\boldsymbol{\mu}\boldsymbol{m}_3^T - \boldsymbol{m}_2\boldsymbol{\nu}^T)\boldsymbol{P}_i = 0, \end{cases} \quad \text{for} \quad i = 1, \dots, n. \tag{1.26}$$

As expected, the vector $\boldsymbol{l}$ associated with $\boldsymbol{\lambda} = \boldsymbol{m}_1$, $\boldsymbol{\mu} = \boldsymbol{m}_2$, and $\boldsymbol{\nu} = \boldsymbol{m}_3$ is a solution of these equations. Are there other solutions?

Let us first consider the case where the points P_i $(i = 1, \dots, n)$ all lie in some plane Π, so $\boldsymbol{P}_i \cdot \boldsymbol{\Pi} = 0$ for some 4-vector $\boldsymbol{\Pi}$. Clearly, choosing $(\boldsymbol{\lambda}, \boldsymbol{\mu}, \boldsymbol{\nu})$ equal to $(\boldsymbol{\Pi}, \mathbf{0}, \mathbf{0})$, $(\mathbf{0}, \boldsymbol{\Pi}, \mathbf{0})$, $(\mathbf{0}, \mathbf{0}, \boldsymbol{\Pi})$, or any linear combination of these vectors will yield a solution of Eq. (1.26). In other words, the nullspace of $\mathcal{P}$ contains the four-dimensional vector space spanned by these vectors and $\boldsymbol{m}$. In practice, this means that the fiducial points P_i should not all lie in the same plane.

In general, for a given nonzero value of the vector $\boldsymbol{l}$, the points P_i that satisfy Eq. (1.26) must lie on the curve where the two quadric surfaces defined by the corresponding equations intersect. A closer look at Eq. (1.26) reveals that the straight line where the planes defined by $\boldsymbol{m}_3 \cdot \boldsymbol{P} = 0$ and $\boldsymbol{\nu} \cdot \boldsymbol{P} = 0$ intersect lies on both quadrics. It can be shown that the intersection curve of these two surfaces consists of this line and of a *twisted cubic* curve Γ passing through the origin. A twisted cubic is determined entirely by six points lying on it, and it follows that seven points chosen at random will not fall on Γ. In addition, since this curve passes through the origin, choosing $n \geq 6$ random points will in general guarantee that the matrix $\mathcal{P}$ has rank 11 and that the projection matrix can be recovered in a unique fashion.

Estimation of the Intrinsic and Extrinsic Parameters. Once the projection matrix $\mathcal{M}$ has been estimated, its expression in terms of the camera's intrinsic and extrinsic parameters (Eq. [1.18]) can be used to recover these parameters as follows: We write as before $\mathcal{M} = \begin{pmatrix} \mathcal{A} & \boldsymbol{b} \end{pmatrix}$, with $\boldsymbol{a}_1^T$, $\boldsymbol{a}_2^T$, and $\boldsymbol{a}_3^T$ denoting the rows of $\mathcal{A}$, and obtain

$$\rho\begin{pmatrix} \mathcal{A} & \boldsymbol{b} \end{pmatrix} = \mathcal{K}\begin{pmatrix} \mathcal{R} & \boldsymbol{t} \end{pmatrix} \Longleftrightarrow \rho\begin{pmatrix} \boldsymbol{a}_1^T \\ \boldsymbol{a}_2^T \\ \boldsymbol{a}_3^T \end{pmatrix} = \begin{pmatrix} \alpha\boldsymbol{r}_1^T - \alpha\cot\theta\boldsymbol{r}_2^T + x_0\boldsymbol{r}_3^T \\ \dfrac{\beta}{\sin\theta}\boldsymbol{r}_2^T + y_0\boldsymbol{r}_3^T \\ \boldsymbol{r}_3^T \end{pmatrix},$$

where ρ is an unknown scale factor, introduced here to account for the fact that the recovered matrix $\mathcal{M}$ has unit Frobenius form since $||\mathcal{M}||_F = ||\boldsymbol{m}|| = 1$.

In particular, using the fact that the rows of a rotation matrix have unit length and are perpendicular to each other yields immediately

$$\begin{cases} \rho = \varepsilon/||\boldsymbol{a}_3||, \\ \boldsymbol{r}_3 = \rho\boldsymbol{a}_3, \\ x_0 = \rho^2(\boldsymbol{a}_1 \cdot \boldsymbol{a}_3), \\ y_0 = \rho^2(\boldsymbol{a}_2 \cdot \boldsymbol{a}_3), \end{cases} \tag{1.27}$$

where $\varepsilon = \mp 1$.

Since θ is always in the neighborhood of $\pi/2$ with a positive sine, we have

$$\begin{cases} \rho^2(\boldsymbol{a}_1 \times \boldsymbol{a}_3) = -\alpha\boldsymbol{r}_2 - \alpha\cot\theta\boldsymbol{r}_1, \\ \rho^2(\boldsymbol{a}_2 \times \boldsymbol{a}_3) = \dfrac{\beta}{\sin\theta}\boldsymbol{r}_1, \end{cases} \quad \text{and} \quad \begin{cases} \rho^2||\boldsymbol{a}_1 \times \boldsymbol{a}_3|| = \dfrac{|\alpha|}{\sin\theta}, \\ \rho^2||\boldsymbol{a}_2 \times \boldsymbol{a}_3|| = \dfrac{|\beta|}{\sin\theta}, \end{cases} \tag{1.28}$$

thus:

$$\begin{cases} \cos\theta = -\dfrac{(\boldsymbol{a}_1 \times \boldsymbol{a}_3)\cdot(\boldsymbol{a}_2 \times \boldsymbol{a}_3)}{||\boldsymbol{a}_1 \times \boldsymbol{a}_3||\,||\boldsymbol{a}_2 \times \boldsymbol{a}_3||}, \\ \alpha = \rho^2||\boldsymbol{a}_1 \times \boldsymbol{a}_3||\sin\theta, \\ \beta = \rho^2||\boldsymbol{a}_2 \times \boldsymbol{a}_3||\sin\theta, \end{cases} \tag{1.29}$$

since the sign of the magnification parameters α and β is normally known in advance and can be taken to be positive.

We can now compute $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ from the second equation in Eq. (1.28) as

$$\begin{cases} \boldsymbol{r}_1 = \dfrac{\rho^2\sin\theta}{\beta}(\boldsymbol{a}_2 \times \boldsymbol{a}_3) = \dfrac{1}{||\boldsymbol{a}_2 \times \boldsymbol{a}_3||}(\boldsymbol{a}_2 \times \boldsymbol{a}_3), \\ \boldsymbol{r}_2 = \boldsymbol{r}_3 \times \boldsymbol{r}_1. \end{cases} \tag{1.30}$$

Note that there are two possible choices for the matrix $\mathcal{R}$, depending on the value of ε. The translation parameters can now be recovered by writing $\mathcal{K}\boldsymbol{t} = \rho\boldsymbol{b}$, and hence $\boldsymbol{t} = \rho\mathcal{K}^{-1}\boldsymbol{b}$. In practical situations, the sign of t_3 is often known in advance (this corresponds to knowing whether the origin of the world coordinate system is in front of or behind the camera), which allows the choice of a unique solution for the calibration parameters.

Figure 1.18 shows the results of an experiment with the dataset from Figure 1.17. The recovered calibration matrix is

$$\mathcal{K} = \begin{pmatrix} 970.2841 & 0.0986 & 372.0050 \\ 0 & 963.3466 & 299.2921 \\ 0 & 0 & 1 \end{pmatrix}$$

for this 768×576 camera, with estimated values of 1.0072 for the aspect ratio, and 0.0058 degree for the skew angle $|\theta - \pi/2|$.[1] The recovered image center is located about 15 pixels away from the center of the image array.

[1] In this book, an $m \times n$ matrix normally has m rows and n columns. Digital images and camera retinas are the only exceptions, and we follow the tradition by assuming that an $m \times n$ picture has m columns and n rows. For example, the camera used in this experiment has 768 columns and 576 rows.

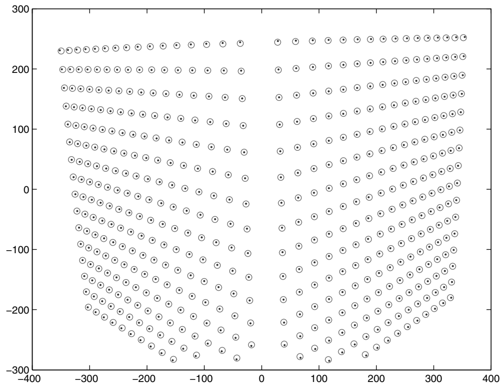

FIGURE 1.18: Results of camera calibration on the dataset shown in Figure 1.17. The original data points (circles) are overlaid with the reprojected 3D points (dots). The root-mean-squared error is 0.96 pixel for this 768×576 image.

1.3.2 A Nonlinear Approach to Camera Calibration

The method presented in the previous section ignores some of the constraints associated with the calibration process. For example, the camera skew was assumed to be arbitrary instead of (very close to) zero in Section 1.3.1. We present in this section a nonlinear approach to camera calibration that takes into account *all* the relevant constraints.

This approach is borrowed from *photogrammetry*, an engineering field whose aim is to recover quantitative geometric information from one or several pictures, with applications in cartography, military intelligence, city planning, etc. For many years, photogrammetry relied on a combination of geometric, optical, and mechanical methods to recover three-dimensional information from pictures, but the advent of computers in the 1950s has made a purely computational approach to this problem feasible. This is the domain of *analytical photogrammetry*, where the intrinsic parameters of a camera define its *interior orientation*, and the extrinsic parameters define its *exterior orientation*.

In this setting, we assume once again that we observe n fiducial points P_i $(i = 1, \dots, n)$ whose positions in some world coordinate system are known, and minimize the mean-squared distance between the measured positions of their images and those predicted by the perspective projection equation with respect to a vector of camera parameters $\boldsymbol{\xi}$ in $\mathbb{R}^{11+q}$, where $q \geq 0$, which might include various distortion coefficients in addition to the usual intrinsic and extrinsic parameters. (This assumes that the aspect-ratio and skew are unknown. When they are known, fewer parameters are necessary.) In particular, let us see how to account for *radial distortion*, a type of aberration that depends on the distance separating the optical axis from the point of interest. We model the projection process by

$$\boldsymbol{p} = \frac{1}{Z}\begin{pmatrix} 1/\lambda & 0 & 0 \\ 0 & 1/\lambda & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathcal{M}\boldsymbol{P}, \tag{1.31}$$

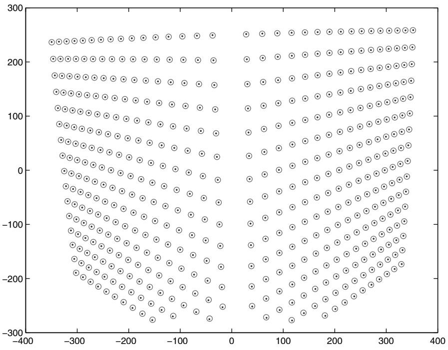

FIGURE 1.19: Results of nonlinear camera calibration on the dataset shown in Figure 1.17. The original data points (circles) are overlaid with the reprojected 3D points (dots). The root-mean-squared error is 0.39 pixel for this 768 × 576 image. Three radial distortion parameters were used in this case.

where λ is a polynomial function of the squared distance between the image center and the image point p in normalized image coordinates, or:

$$d^2 = \hat{x}^2 + \hat{y}^2 = ||\mathcal{K}^{-1}\boldsymbol{p}||^2 - 1. \tag{1.32}$$

In most applications, it is sufficient to use a low-degree polynomial (e.g., $\lambda = 1 + \sum_{p=1}^{q} \kappa_p d^{2p}$, with $q \leq 3$) and the *distortion coefficients* κ_p ($p = 1, \ldots, q$) are normally assumed to be small.

Using Eq. (1.32) to write λ as an explicit function of $\boldsymbol{p}$ in Eq. (1.31) yields highly nonlinear constraints on the $11 + q$ camera parameters. The least-squares error can be written as

$$E(\boldsymbol{\xi}) = \sum_{i=1}^{2n} f_i^2(\boldsymbol{\xi}), \text{ where } \begin{cases} f_{2j-1}(\boldsymbol{\xi}) & = & x_j - \dfrac{\boldsymbol{m}_1(\boldsymbol{\xi}) \cdot \boldsymbol{P}_j}{\boldsymbol{m}_3(\boldsymbol{\xi}) \cdot \boldsymbol{P}_j}, \\ f_{2j}(\boldsymbol{\xi}) & = & y_j - \dfrac{\boldsymbol{m}_2(\boldsymbol{\xi}) \cdot \boldsymbol{P}_j}{\boldsymbol{m}_3(\boldsymbol{\xi}) \cdot \boldsymbol{P}_j}, \end{cases} \text{ for } j = 1, \ldots, n. \tag{1.33}$$

Contrary to the cases studied so far, the dependency of each error term $f_i(\boldsymbol{\xi})$ on the unknown parameters $\boldsymbol{\xi}$ is not linear. Instead, it involves a combination of polynomial and trigonometric functions, and minimizing the overall error measure involves the use of the nonlinear least-squares algorithms discussed in Chapter 22. These algorithms require computing the Jacobian of the vector function $\boldsymbol{f}(\boldsymbol{\xi}) = (f_1[\boldsymbol{\xi}], \ldots, f_{2n}[\boldsymbol{\xi}])^T$ with respect to the vector $\boldsymbol{\xi}$ of unknown parameters, which is easily done analytically (see problems).

Figure 1.19 shows the results of an experiment with the dataset from Figure 1.17 using three radial distortion coefficients. The recovered calibration matrix is

$$\mathcal{K} = \begin{pmatrix} 1014.0 & 0.0001 & 371.8 \\ 0 & 1008.9 & 292.3 \\ 0 & 0 & 1 \end{pmatrix}$$

Perspective projection	$\begin{cases} x = d\dfrac{X}{Z} \\ y = d\dfrac{Y}{Z} \end{cases}$	X, Y, Z: world coordinates ($Z < 0$) x, y: image coordinates d: pinhole-to-retina distance
Weak-perspective projection	$\begin{cases} x' = -mX \\ y' = -mY \\ m = -\dfrac{d}{Z_0} \end{cases}$	X, Y: world coordinates x, y: image coordinates d: pinhole-to-retina distance Z_0: reference-point depth (< 0) m: magnification (> 0)
Orthographic projection	$\begin{cases} x = X \\ y = Y \end{cases}$	X, Y: world coordinates x, y: image coordinates
Thin lens equation	$\dfrac{1}{z} - \dfrac{1}{Z} = \dfrac{1}{f}$	Z: object-point depth (< 0) z: image-point depth (> 0) f: focal length

TABLE 1.1: Reference card: Projection models.

for this 768×576 camera, with estimated values of 1.0051 for the aspect ratio, and less than 10^-5 degree for the skew angle. The recovered image center is located about 9 pixels away from the center of the image array. The three radial distortion coefficients are -0.1183, -0.3657, and 1.9112 in this case.

1.4 NOTES

The classical textbook by Hecht (1987) is an excellent introduction to geometric optics, paraxial refraction, thin and thick lenses, and their aberrations, as briefly discussed in Section 1.1. Vignetting is discussed in Horn (1986). Wandell (1995) gives an excellent treatment of image formation in the human visual system. Thorough presentations of the geometric camera models discussed in Section 1.2 can be found in Faugeras (1993), Hartley and Zisserman (2000*b*), and Faugeras *et al.* (2001). The paraperspective projection model was introduced in computer vision by Ohta, Maenobu, and Sakai (1981), and its properties have been studied by Aloimonos (1990).

The linear calibration technique described in Section 1.3.1 is detailed in Faugeras (1993). Its variant that takes radial distortion into account is adapted from Tsai (1987). The book of Haralick and Shapiro (1992) presents a concise introduction to analytical photogrammetry. The *Manual of Photogrammetry* is of course the gold standard, and newcomers to this field (like the authors of this book) will probably find the ingenious mechanisms and rigorous methods described in the various editions of this book fascinating (Thompson *et al.* 1966; Slama *et al.* 1980). We will come back to photogrammetry in the context of structure from motion in Chapter 8. The linear and nonlinear least-squares techniques used in the approaches to camera calibration discussed in the present chapter are presented in some detail in Chapter 22. An excellent survey and discussion of these methods in the context of analytical photogrammetry can be found in Triggs *et al.* (2000).

We have assumed in this chapter that a 3D calibration rig is available. This is

Perspective projection equation (homogeneous)	$\boldsymbol{p} = \frac{1}{Z}\mathcal{M}\boldsymbol{P}$
Matrix of intrinsic parameters	$\mathcal{K} = \begin{pmatrix} \alpha & -\alpha\cot\theta & x_0 \\ 0 & \beta/\sin\theta & y_0 \\ 0 & 0 & 1 \end{pmatrix}$
Perspective projection matrix	$\mathcal{M} = \mathcal{K}\begin{pmatrix}\mathcal{R} & \boldsymbol{t}\end{pmatrix}$
Affine projection equation (nonhomogeneous)	$\boldsymbol{p} = \mathcal{M}\begin{pmatrix}\boldsymbol{P} \\ 1\end{pmatrix} = \mathcal{A}\boldsymbol{P} + \boldsymbol{b}$
Weak-perspective projection matrix	$\mathcal{M} = \begin{pmatrix}\mathcal{A} & \boldsymbol{b}\end{pmatrix} = \frac{1}{Z_r}\begin{pmatrix} k & s \\ 0 & 1\end{pmatrix}\begin{pmatrix}\mathcal{R}_2 & \boldsymbol{t}_2\end{pmatrix}$

TABLE 1.2: Reference card: Geometric camera models.

the setting used in (Faig 1975; Tsai 1987; Faugeras 1993; Heikkilä 2000) for example. However, it is difficult to build such a rig accurately—see Lavest, Viala, and Dhome (1998) for a discussion of this problem and an ingenious solution—and many authors prefer using multiple checkerboards or similar planar patterns (Devy, Garric & Orteu 1997; Zhang 2000). This includes the widely used C implementation of J.-Y. Bouguet's algorithm, distributed as part of *OpenCV*, an open-source library of computer vision routines, available at `http://opencv.willowgarage.com/wiki/`. A MATLAB version is also freely available at his web site; see: `http://www.vision.caltech.edu/bouguetj/calib_doc/`.

Given the fundamental importance of the notions introduced in this chapter, the main equations derived in its course have been collected in Tables 1.1 and 1.2 for reference.

PROBLEMS

1.1. Derive the perspective equation projections for a virtual image located at a distance d *in front* of the pinhole.

1.2. Prove geometrically that the projections of two parallel lines lying in some plane Φ appear to converge on a horizon line h formed by the intersection of the image plane Π with the plane parallel to Φ and passing through the pinhole.

1.3. Prove the same result algebraically using the perspective projection Eq. (1.1). You can assume for simplicity that the plane Φ is orthogonal to the image plane Π.

1.4. Consider a camera equipped with a thin lens, with its image plane at position z and the plane of scene points in focus at position Z. Now suppose that the image plane is moved to $\hat{z}$. Show that the diameter of the corresponding blur circle is

$$a\frac{|z - \hat{z}|}{z},$$

where a is the lens diameter. Use this result to show that the depth of field (i.e., the distance between the near and far planes that will keep the diameter of the blur circles below some threshold ε) is given by

$$D = 2\varepsilon f Z(Z+f)\frac{a}{f^2a^2 - \varepsilon^2 Z^2},$$

and conclude that, for a *fixed* focal length, the depth of field increases as the lens diameter decreases, and thus the f number f/a increases.
Hint: Solve for the depth $\hat{Z}$ of a point whose image is focused on the image plane at position $\hat{z}$, considering both the case where $\hat{z}$ is larger than z and the case where it is smaller.

1.5. Give a geometric construction of the image p of a point P given the two focal points F and F' of a thin lens.

1.6. Show that when the camera coordinate system is skewed and the angle θ between the two image axes is not equal to 90 degrees, then Eq. (1.12) transforms into Eq. (1.13).

1.7. Consider two Euclidean coordinate systems $(A) = (O_A, \boldsymbol{i}_A, \boldsymbol{j}_A, \boldsymbol{k}_A)$ and $(B) = (O_B, \boldsymbol{i}_B, \boldsymbol{j}_B, \boldsymbol{k}_B)$. Show that Eq. (1.8) follows from Eq. (1.6).
Hint: First consider the case where both coordinate systems share the same basis vectors—that is, $\boldsymbol{i}_A = \boldsymbol{i}_B$, $\boldsymbol{j}_A = \boldsymbol{j}_B$, and $\boldsymbol{k}_A = \boldsymbol{k}_B$. Then consider the case where they have the same origin—that is, $O_A = O_B$, before finally treating the general case.

1.8. Write formulas for the matrices $\mathcal{R}$ in Eq. (1.8) when (B) is deduced from (A) via a rotation of angle θ about the axes $\boldsymbol{i}_A$, $\boldsymbol{j}_A$, and $\boldsymbol{k}_A$ respectively.

1.9. Let $\boldsymbol{O}$ denote the *homogeneous* coordinate vector of the optical center of a camera in some reference frame, and let $\mathcal{M}$ denote the corresponding perspective projection matrix. Show that $\mathcal{M}\boldsymbol{O} = \mathbf{0}$. Explain why this intuitively makes sense.

1.10. Show that the conditions of Theorem 1 are necessary.

1.11. Show that any affine projection matrix $\mathcal{M} = \begin{pmatrix} \mathcal{A} & \boldsymbol{b} \end{pmatrix}$ can be written as a general weak-perspective projection matrix as defined by Eq. (1.22), i.e.,

$$\mathcal{M} = \frac{1}{Z_r}\begin{pmatrix} k & s \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \mathcal{R}_2 & \boldsymbol{t}_2 \end{pmatrix}.$$

1.12. Give an analytical expression for the Jacobian of the vector function $\boldsymbol{f}(\boldsymbol{\xi}) = (f_1[\boldsymbol{\xi}], \dots, f_{2n}[\boldsymbol{\xi}])^T$ featured in Eq. (1.33) with respect to the vector $\boldsymbol{\xi}$ of unknown parameters.

PROGRAMMING EXERCISES

1.13. Implement the linear calibration algorithm presented in Section 1.3.1.

1.14. Implement the nonlinear calibration algorithm from Section 1.3.2.

CHAPTER 2

Light and Shading

The brightness of a pixel in the image is a function of the brightness of the surface patch in the scene that projects to the pixel. In turn, the brightness of the patch depends on how much incident light arrives at the patch and on the fraction of the incident light that gets reflected (Models in Section 2.1).

This means that the brightness of a pixel is profoundly ambiguous. Surprisingly, people can disentangle these effects quite accurately. Often, but not always, people can tell whether objects are in bright light or in shadow, and do not perceive objects in shadow as having dark surfaces. People can usually tell whether changes of brightness are caused by changes in reflection or by shading (cinemas wouldn't work if we got it right all the time, however). Typically, people can tell that shading comes from the geometry of the object, but sometimes get shading and markings mixed up. For example, a streak of dark makeup under a cheekbone will often look like a shading effect, making the face look thinner. Quite simple models of shading (Section 2.1) support a range of inference procedures (Section 2.2). More complex models are needed to explain some important effects (Section 2.1.4), but make inference very difficult indeed (Section 2.4).

2.1 MODELLING PIXEL BRIGHTNESS

Three major phenomena determine the brightness of a pixel: the response of the camera to light, the fraction of light reflected from the surface to the camera, and the amount of light falling on the surface. Each can be dealt with quite straightforwardly.

Camera response: Modern cameras respond linearly to middling intensities of light, but have pronounced nonlinearities for darker and brighter illumination. This allows the camera to reproduce the very wide dynamic range of natural light without saturating. For most purposes, it is enough to assume that the camera response is linearly related to the intensity of the surface patch. Write $\boldsymbol{X}$ for a point in space that projects to $\boldsymbol{x}$ in the image, $I_{patch}(\boldsymbol{X})$ for the intensity of the surface patch at $\boldsymbol{X}$, and $I_{camera}(\boldsymbol{x})$ for the camera response at $\boldsymbol{x}$. Then our model is:

$$I_{camera}(\boldsymbol{x}) = kI_{patch}(\boldsymbol{x}),$$

where k is some constant to be determined by calibration. Generally, we assume that this model applies and that k is known if needed. Under some circumstances, a more complex model is appropriate; we discuss how to recover such models in Section 2.2.1.

Surface reflection: Different points on a surface may reflect more or less of the light that is arriving. Darker surfaces reflect less light, and lighter surfaces reflect more. There is a rich set of possible physical effects, but most can be ignored. Section 2.1.1 describes the relatively simple model that is sufficient for almost all

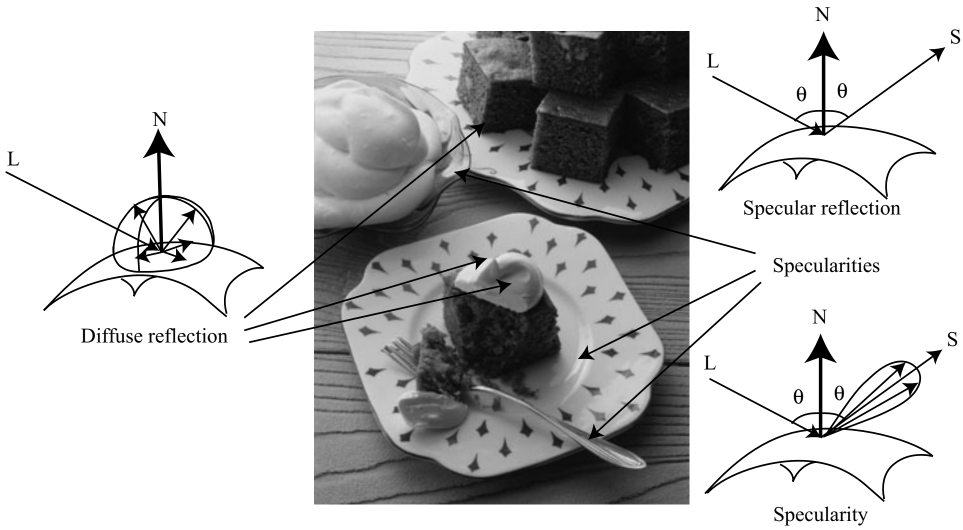

FIGURE 2.1: The two most important reflection modes for computer vision are diffuse reflection (**left**), where incident light is spread evenly over the whole hemisphere of outgoing directions, and specular reflection (**right**), where reflected light is concentrated in a single direction. The specular direction $\boldsymbol{S}$ is coplanar with the normal and the source direction ($\boldsymbol{L}$), and has the same angle to the normal that the source direction does. Most surfaces display both diffuse and specular reflection components. In most cases, the specular component is not precisely mirror like, but is concentrated around a range of directions close to the specular direction (**lower right**). This causes specularities, where one sees a mirror like reflection of the light source. Specularities, when they occur, tend to be small and bright. In the photograph, they appear on the metal spoon and on the plate. Large specularities can appear on flat metal surfaces (arrows). Most curved surfaces (such as the plate) show smaller specularities. Most of the reflection here is diffuse; some cases are indicated by arrows. *Martin Brigdale © Dorling Kindersley, used with permission.*

purposes in computer vision.

Illumination: The amount of light a patch receives depends on the overall intensity of the light, and on the geometry. The overall intensity could change because some *luminaires* (the formal term for light sources) might be shadowed, or might have strong directional components. Geometry affects the amount of light arriving at a patch because surface patches facing the light collect more radiation and so are brighter than surface patches tilted away from the light, an effect known as *shading*. Section 2.1.2 describes the most important model used in computer vision; Section 2.3 describes a much more complex model that is necessary to explain some important practical difficulties in shading inference.

2.1.1 Reflection at Surfaces

Most surfaces reflect light by a process of *diffuse reflection*. Diffuse reflection scatters light evenly across the directions leaving a surface, so the brightness of a diffuse surface doesn't depend on the viewing direction. Examples are easy to identify with this test: most cloth has this property, as do most paints, rough wooden surfaces, most vegetation, and rough stone or concrete. The only parameter required to describe a surface of this type is its *albedo*, the fraction of the light arriving at the surface that is reflected. This does not depend on the direction in which the light arrives or the direction in which the light leaves. Surfaces with very high or very low albedo are difficult to make. For practical surfaces, albedo lies in the range $0.05 - 0.90$ (see Brelstaff and Blake (1988*b*), who argue the dynamic range is closer to 10 than the 18 implied by these numbers). Mirrors are not diffuse, because what you see depends on the direction in which you look at the mirror. The behavior of a perfect mirror is known as *specular reflection*. For an ideal mirror, light arriving along a particular direction can leave only along the *specular direction*, obtained by reflecting the direction of incoming radiation about the surface normal (Figure 2.1). Usually some fraction of incoming radiation is absorbed; on an ideal specular surface, this fraction does not depend on the incident direction.

If a surface behaves like an ideal specular reflector, you could use it as a mirror, and based on this test, relatively few surfaces actually behave like ideal specular reflectors. Imagine a near perfect mirror made of polished metal; if this surface suffers slight damage at a small scale, then around each point there will be a set of small facets, pointing in a range of directions. In turn, this means that light arriving in one direction will leave in several different directions because it strikes several facets, and so the specular reflections will be blurred. As the surface becomes less flat, these distortions will become more pronounced; eventually, the only specular reflection that is bright enough to see will come from the light source. This mechanism means that, in most shiny paint, plastic, wet, or brushed metal surfaces, one sees a bright blob—often called a *specularity*—along the specular direction from light sources, but few other specular effects. Specularities are easy to identify, because they are small and very bright (Figure 2.1; Brelstaff and Blake (1988*b*)). Most surfaces reflect only some of the incoming light in a specular component, and we can represent the percentage of light that is specularly reflected with a *specular albedo*. Although the diffuse albedo is an important material property that we will try to estimate from images, the specular albedo is largely seen as a nuisance and usually is not estimated.

2.1.2 Sources and Their Effects

The main source of illumination outdoors is the sun, whose rays all travel parallel to one another in a known direction because it is so far away. We model this behavior with a *distant point light source*. This is the most important model of lighting (because it is like the sun and because it is easy to use), and can be quite effective for indoor scenes as well as outdoor scenes. Because the rays are parallel to one another, a surface that faces the source cuts more rays (and so collects more light) than one oriented along the direction in which the rays travel. The amount of light collected by a surface patch in this model is proportional to the cosine of the angle θ between the illumination direction and the normal (Figure 2.2). The figure yields *Lambert's cosine law*, which states the brightness of a diffuse patch illuminated by a distant point light source is given by

$$I = \rho I_0 \cos\theta,$$

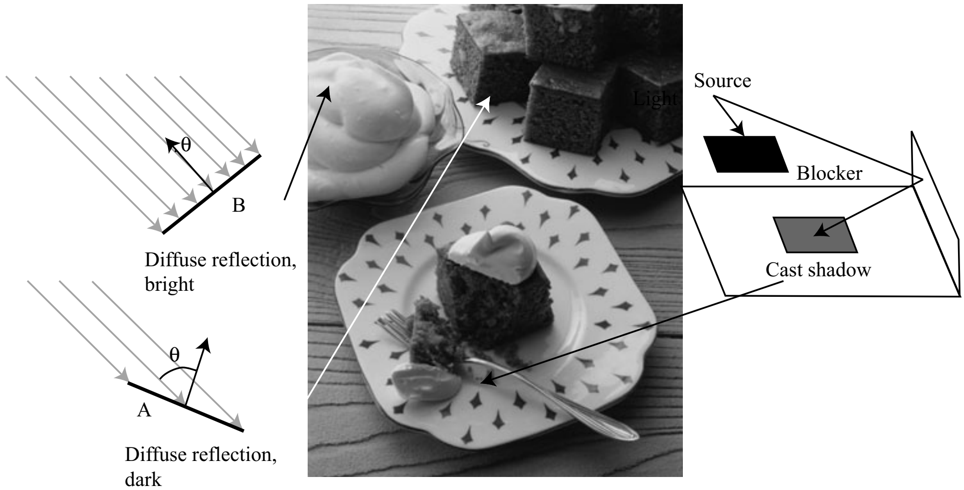

FIGURE 2.2: The orientation of a surface patch with respect to the light affects how much light the patch gathers. We model surface patches as illuminated by a distant point source, whose rays are shown as light arrowheads. Patch A is tilted away from the source (θ is close to 90^0) and collects less energy, because it cuts fewer light rays per unit surface area. Patch B, facing the source (θ is close to 0^0), collects more energy, and so is brighter. Shadows occur when a patch cannot see a source. The shadows are not dead black, because the surface can see interreflected light from other surfaces. These effects are shown in the photograph. The darker surfaces are turned away from the illumination direction. *Martin Brigdale © Dorling Kindersley, used with permission.*

where I_0 is the intensity of the light source, θ is the angle between the light source direction and the surface normal, and ρ is the diffuse albedo. This law predicts that bright image pixels come from surface patches that face the light directly and dark pixels come from patches that see the light only tangentially, so that the shading on a surface provides some shape information. We explore this cue in Section 2.4.

If the surface cannot see the source, then it is in *shadow*. Since we assume that light arrives at our patch only from the distant point light source, our model suggests that shadows are deep black; in practice, they very seldom are, because the shadowed surface usually receives light from other sources. Outdoors, the most important such source is the sky, which is quite bright. Indoors, light reflected from other surfaces illuminates shadowed patches. This means that, for example, we tend to see few shadows in rooms with white walls, because any shadowed patch receives a lot of light from the walls. These *interreflections* also can have a significant effect on the brightness surfaces that are not in shadow. Interreflection effects are sometimes modelled by adding a constant *ambient illumination* term to the predicted intensity. The ambient term ensures that shadows are not too dark, but this is not a particularly successful model of the spatial properties of interreflections. More detailed models require some familiarity with radiometric terminology, but they are important in some applications; we have confined this topic to Section 2.3.

2.1.3 The Lambertian+Specular Model

For almost all purposes, it is enough to model all surfaces as being diffuse with specularities. This is the *lambertian+specular model*. Specularities are relatively seldom used in inference (Section 2.2.2 sketches two methods), and so there is no need for a formal model of their structure. Because specularities are small and bright, they are relatively easy to identify and remove with straightforward methods (find small bright spots, and replace them by smoothing the local pixel values). More sophisticated specularity finders use color information (Section 3.5.1). Thus, to apply the lambertian+specular model, we find and remove specularities, and then use Lambert's law (Section 2.1.2) to model image intensity.

We must choose which source effects to model. In the simplest case, a *local shading model*, we assume that shading is caused only by light that comes from the luminaire (i.e., that there are no interreflections).

Now assume that the luminaire is an infinitely distant source. For this case, write $\boldsymbol{N}(x)$ for the unit surface normal at $\boldsymbol{x}$, $\boldsymbol{S}$ for a vector pointing from $\boldsymbol{x}$ toward the source with length I_o (the source intensity), $\rho(\boldsymbol{x})$ for the albedo at $\boldsymbol{x}$, and $Vis(\boldsymbol{S}, \boldsymbol{x})$ for a function that is 1 when $\boldsymbol{x}$ can see the source and zero otherwise. Then, the intensity at $\boldsymbol{x}$ is

$$\begin{array}{ccccccc} I(\boldsymbol{x}) & = & \rho(\boldsymbol{x})\,(\boldsymbol{N}\cdot\boldsymbol{S})\,\mathrm{Vis}(\boldsymbol{S},\,\boldsymbol{x}) & + & \rho(\boldsymbol{x})A & + & M \\ & & & & & & \\ \text{Image} & = & \text{Diffuse} & + & \text{Ambient} & + & \text{Specular (mirror-like)} \\ \text{intensity} & & \text{term} & & \text{term} & & \text{term} \end{array}$$

This model can still be used for a more complex source (for example, an area source), but in that case it is more difficult to determine an appropriate $\boldsymbol{S}(\boldsymbol{x})$.

2.1.4 Area Sources

An *area source* is an area that radiates light. Area sources occur quite commonly in natural scenes—an overcast sky is a good example—and in synthetic environments—for example, the fluorescent light boxes found in many industrial ceilings. Area sources are common in illumination engineering, because they tend not to cast strong shadows and because the illumination due to the source does not fall off significantly as a function of the distance to the source. Detailed models of area sources are complex (Section 2.3), but a simple model is useful to understand shadows. Shadows from area sources are very different from shadows cast by point sources. One seldom sees dark shadows with crisp boundaries indoors. Instead, one could see no visible shadows, or shadows that are rather fuzzy diffuse blobs, or sometimes fuzzy blobs with a dark core (Figure 2.3). These effects occur because rooms tend to have light walls and diffuse ceiling fixtures, which act as area sources. As a result, the shadows one sees are area source shadows.

To compute the intensity at a surface patch illuminated by an area source, we can break the source up into infinitesimal source elements, then sum effects from each element. If there is an occluder, then some surface patches may see none of the source elements. Such patches will be dark, and lie in the *umbra* (a Latin word meaning "shadow"). Other surface patches may see some, but not all, of the source elements. Such patches may be quite bright (if they see most of the elements), or

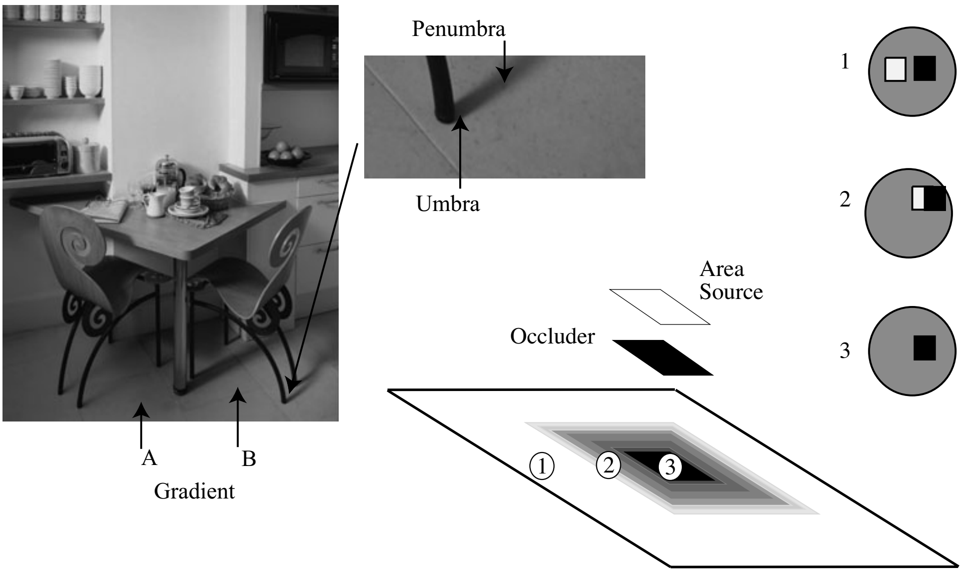

FIGURE 2.3: Area sources generate complex shadows with smooth boundaries, because from the point of view of a surface patch, the source disappears slowly behind the occluder. **Left:** a photograph, showing characteristic area source shadow effects. Notice that A is much darker than B; there must be some shadowing effect here, but there is no clear shadow boundary. Instead, there is a fairly smooth gradient. The chair leg casts a complex shadow, with two distinct regions. There is a core of darkness (the *umbra*—where the source cannot be seen at all) surrounded by a partial shadow (*penumbra*—where the source can be seen partially). A good model of the geometry, illustrated **right**, is to imagine lying with your back to the surface looking at the world above. At point 1, you can see all of the source; at point 2, you can see some of it; and at point 3, you can see none of it. *Peter Anderson © Dorling Kindersley, used with permission.*

relatively dark (if they see few elements), and lie in the *penumbra* (a compound of Latin words meaning "almost shadow"). One way to build intuition is to think of a tiny observer looking up from the surface patch. At umbral points, this observer will not see the area source at all whereas at penumbral points, the observer will see some, but not all, of the area source. An observer moving from outside the shadow, through the penumbra and into the umbra will see something that looks like an eclipse of the moon (Figure 2.3). The penumbra can be large, and can change quite slowly from light to dark. There might even be no umbral points at all, and, if the occluder is sufficiently far away from the surface, the penumbra could be very large and almost indistinguishable in brightness from the unshadowed patches. This is why many objects in rooms appear to cast no shadow at all (Figure 2.4).

2.2 INFERENCE FROM SHADING

Shading can be used to infer a variety of properties of the visual world. Successful inference often requires that we calibrated the camera radiometrically, so that we know how pixel values map to radiometric values (Section 2.2.1). As Figure 2.1

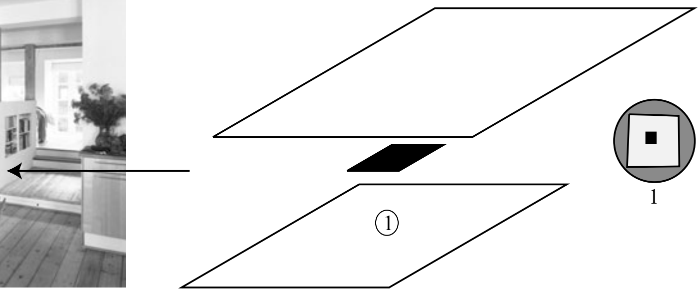

FIGURE 2.4: The photograph on the **left** shows a room interior. Notice the lighting has some directional component (the vertical face indicated by the arrow is dark, because it does not face the main direction of lighting), but there are few visible shadows (for example, the chairs do not cast a shadow on the floor). On the **right**, a drawing to show why; here there is a small occluder and a large area source. The occluder is some way away from the shaded surface. Generally, at points on the shaded surface the incoming hemisphere looks like that at point 1. The occluder blocks out some small percentage of the area source, but the amount of light lost is too small to notice (compare figure 2.3). *Jake Fitzjones © Dorling Kindersley, used with permission.*

suggests, specularities are a source of information about the shape of a surface, and Section 2.2.2 shows how this information can be interpreted. Section 2.2.3 shows how to recover the albedoes of surfaces from images. Finally, Section 2.2.4 shows how multiple shaded images can be used to recover surface shape.

2.2.1 Radiometric Calibration and High Dynamic Range Images

Real scenes often display a much larger range of intensities than cameras can cope with. Film and charge-coupled devices respond to energy. A property called *reciprocity* means that, if a scene patch casts intensity E onto the film, and if the shutter is open for time Δt, the response is a function of $E\Delta t$ alone. In particular, we will get the same outcome if we image one patch of intensity E for time Δt and another patch of intensity E/k for time $k\Delta t$. The actual response that the film produces is a function of $E\Delta t$; this function might depend on the imaging system, but is typically somewhat linear over some range, and sharply non-linear near the top and bottom of this range, so that the image can capture very dark and very light patches without saturation. It is usually monotonically increasing.

There are a variety of applications where it would be useful to know the actual radiance (equivalently, the intensity) arriving at the imaging device. For example, we might want to compare renderings of a scene with pictures of the scene, and to do that we need to work in real radiometric units. We might want to use pictures of a scene to estimate the lighting in that scene so we can postrender new objects into the scene, which would need to be lit correctly. To infer radiance, we must determine the film response, a procedure known as *radiometric calibration*. As we shall see, doing this will require more than one image of a scene, each obtained at different exposure settings. Imagine we are looking at a scene of a stained glass window lit from behind in a church. At one exposure setting, we would be able to resolve detail in the dark corners, but not on the stained glass, which would be saturated. At another setting, we would be able to resolve detail on the glass, but the interior would be too dark. If we have both settings, we may as well try to recover radiance with a very large dynamic range—producing a *high dynamic range image*.

Now assume we have multiple registered images, each obtained using a different exposure time. At the i, j'th pixel, we know the image intensity value $I_{ij}^{(k)}$ for the k'th exposure time, we know the value of the k'th exposure time Δt_k, and we know that the intensity of the corresponding surface patch E_{ij} is the same for each exposure, but we do not know the value of E_{ij}. Write the camera response function f, so that

$$I_{ij}^{(k)} = f(E_{ij}\Delta t_k).$$

There are now several possible approaches to solve for f. We could assume a parametric form—say, polynomial—then solve using least squares. Notice that we must solve not only for the parameters of f, but also for E_{ij}. For a color camera, we solve for calibration of each channel separately. Mitsunaga and Nayar (1999) have studied the polynomial case in detail. Though the solution is not unique, ambiguous solutions are strongly different from one another, and most cases are easily ruled out. Furthermore, one does not need to know exposure times with exact accuracy to estimate a solution, as long as there are sufficient pixel values; instead, one estimates f from a fixed set of exposure times, then estimates the exposure times from f, and then re-estimates. This procedure is stable.

Alternatively, because the camera response is monotonic, we can work with its inverse $g = f^{-1}$, take logs, and write

$$\log g(I_{ij}^{(k)}) = \log E_{ij} + \log \Delta t_k.$$

We can now estimate the values that g takes at each point and the E_{ij} by placing a smoothness penalty on g. In particular, we minimize

$$\sum_{i,j,k} (\log g(I_{ij}^{(k)}) - (\log E_{ij} + \log \Delta t_k))^2 + \text{smoothness penalty on } g$$

by choice of g. Debevec and Malik (1997) penalize the second derivative of g. Once we have a radiometrically calibrated camera, estimating a high dynamic range image is relatively straightforward. We have a set of registered images, and at each pixel location, we seek the estimate of radiance that predicts the registered image values best. In particular, we assume we know f. We seek an E_{ij} such that

$$\sum_k w(I_{ij})(I_{ij}^{(k)} - f(E_{ij}\Delta t_k))^2$$

is minimized. Notice the weights because our estimate of f is more reliable when I_{ij} is in the middle of the available range of values than when it is at larger or smaller values.

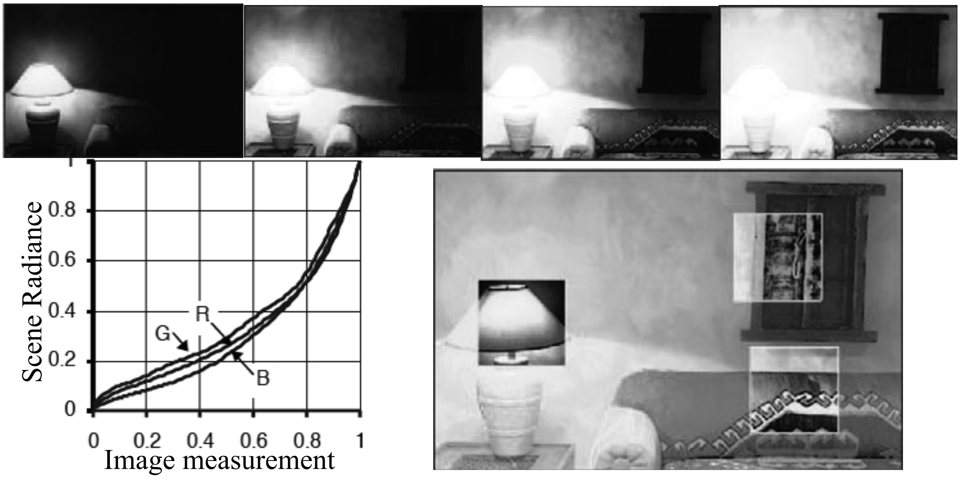

FIGURE 2.5: It is possible to calibrate the radiometric response of a camera from multiple images obtained at different exposures. The **top** row shows four different exposures of the same scene, ranging from darker (shorter shutter time) to lighter (longer shutter time). Note how, in the dark frames, the lighter part of the image shows detail, and in the light frames, the darker part of the image shows detail; this is the result of non-linearities in the camera response. On the **bottom left**, we show the inferred calibration curves for each of the R, G, and B camera channels. On the **bottom right**, a composite image illustrates the results. The dynamic range of this image is far too large to print; instead, the main image is normalized to the print range. Overlaid on this image are boxes where the radiances in the box have also been normalized to the print range; these show how much information is packed into the high dynamic range image. *This figure was originally published as Figure 7 of "Radiometric Self Calibration," by T. Mitsunaga and S. Nayar, Proc. IEEE CVPR 1999, © IEEE, 1999.*

2.2.2 The Shape of Specularities

Specularities are informative. They offer hints about the color of illumination (see Chapter 3) and offer cues to the local geometry of a surface. Understanding these cues is a useful exercise in differential geometry. We consider a smooth specular surface and assume that the brightness reflected in the direction $\boldsymbol{V}$ is a function of $\boldsymbol{V} \cdot \boldsymbol{P}$, where $\boldsymbol{P}$ is the specular direction. We expect the specularity to be small and isolated, so we can assume that the source direction $\boldsymbol{S}$ and the viewing direction $\boldsymbol{V}$ are constant over its extent. Let us further assume that the specularity can be defined by a threshold on the specular energy, i.e., $\boldsymbol{V} \cdot \boldsymbol{P} \geq 1 - \varepsilon$ for some constant ε, denote by $\boldsymbol{N}$ the unit surface normal, and define the *half-angle direction* as $\boldsymbol{H} = (\boldsymbol{S} + \boldsymbol{V})/2$ (Figure 2.6(left)). Using the fact that the vectors $\boldsymbol{S}$, $\boldsymbol{V}$ and $\boldsymbol{P}$ have unit length and a whit of plane geometry, it can easily be shown that the boundary of the specularity is defined by (see exercises)

$$1 - \varepsilon = \boldsymbol{V} \cdot \boldsymbol{P} = 2\frac{(\boldsymbol{H} \cdot \boldsymbol{N})^2}{(\boldsymbol{H} \cdot \boldsymbol{H})} - 1. \tag{2.1}$$

Because the specularity is small, the second-order structure of the surface

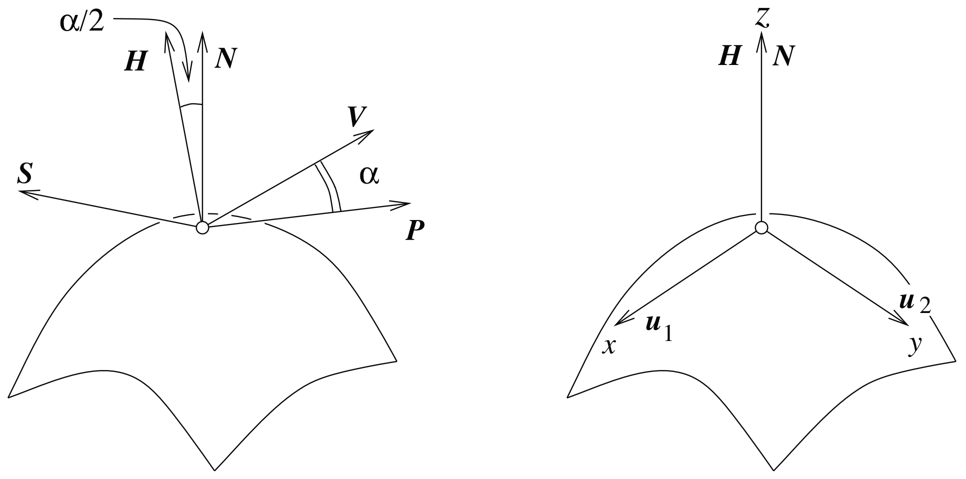

FIGURE 2.6: A specular surface viewed by a distant observer. We establish a coordinate system at the brightest point of the specularity (where the half-angle direction is equal to the normal) and orient the system using the normal and principal directions.

will allow us to characterize the shape of its boundary as follows: there is some point on the surface inside the specularity (in fact, the brightest point) where $\boldsymbol{H}$ is parallel to $\boldsymbol{N}$. We set up a coordinate system at this point, oriented so that the z-axis lies along $\boldsymbol{N}$ and the x- and y-axes lie along the principal directions $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ (Figure 2.6(right)). As noted earlier, the surface can be represented up to second order as $z = -1/2(\kappa_1 x^2 + \kappa_2 y^2)$ in this frame, where κ_1 and κ_2 are the principal curvatures. Now, let us define a *parametric surface* as a differentiable mapping $\boldsymbol{x} : U \subset \mathbb{R}^2 \to \mathbb{R}^3$ associating with any couple $(u, v) \in U$ the coordinate vector $(x, y, z)^T$ of a point in some fixed coordinate system. It is easily shown (see exercises) that the normal to a parametric surface is along the vector $\frac{\partial}{\partial u}\boldsymbol{x} \times \frac{\partial}{\partial v}\boldsymbol{x}$. Our second-order surface model is a parametric surface parameterized by x and y, thus its unit surface normal is defined in the corresponding frame by

$$\boldsymbol{N}(x, y) = \frac{1}{\sqrt{1 + \kappa_1^2 x^2 + \kappa_2^2 y^2}} \begin{pmatrix} \kappa_1 x \\ \kappa_2 y \\ 1 \end{pmatrix},$$

and $\boldsymbol{H} = (0, 0, 1)^T$. Because $\boldsymbol{H}$ is a constant, we can rewrite Eq. (2.1) as $\kappa_1^2 x^2 + \kappa_2^2 y^2 = \zeta$, where ζ is a constant depending on ε. In particular, the shape of the specularity on the surface contains information about the second fundamental form. The specularity will be an ellipse, with major and minor axes oriented along the principal directions, and an eccentricity equal to the ratio of the principal curvatures. Unfortunately, the shape of the specularity on the surface is not, in general, directly observable, so this property can be exploited only when a fair amount about the viewing and illumination setup is known (Healey and Binford 1986).

Although we cannot get much out of the shape of the specularity in the image, it is possible to tell a convex surface from a concave one by watching how a specularity moves as the view changes (you can convince yourself of this with the aid of a spoon).[1] Let us consider a point source at infinity and assume that the specular lobe is very narrow, so the viewing direction and the specular direction coincide. Initially, the specular direction is $\boldsymbol{V}$, and the specularity is at the surface point P; after a small eye motion, $\boldsymbol{V}$ changes to $\boldsymbol{V}'$, while the specularity moves to the close-by point P' (Figure 2.7).

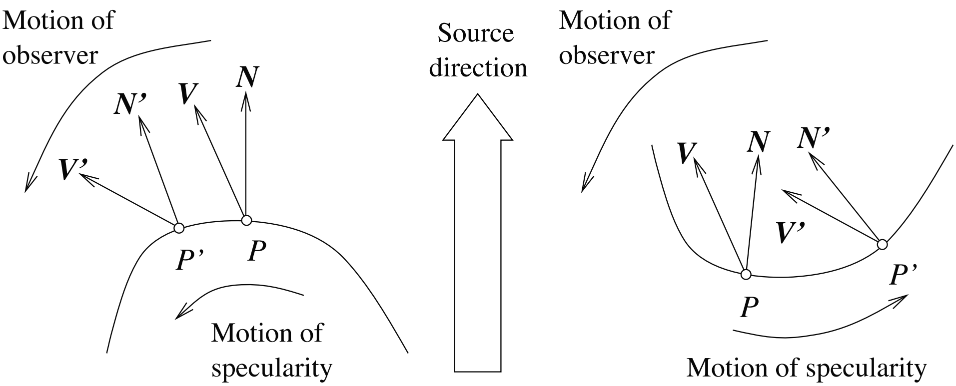

FIGURE 2.7: Specularities on convex and concave surfaces behave differently when the view changes. With an appropriate choice of source direction and motion, this could be used to obtain the signs of the principal curvatures.

The quantity of interest is $\delta a = (\boldsymbol{V}' - \boldsymbol{V}) \cdot \boldsymbol{t}$, where $\boldsymbol{t} = \frac{1}{\delta s}\overrightarrow{PP'}$ is tangent to the surface, and δs is the (small) distance between P and P': if δa is positive, then the specularity moves in the direction of the view (back of the spoon), and if it is negative, the specularity moves against the direction of the view (bowl of the spoon). By construction, we have $\boldsymbol{V} = 2(\boldsymbol{S} \cdot \boldsymbol{N})\boldsymbol{N} - \boldsymbol{S}$, and

$$\begin{aligned} \boldsymbol{V}' &= 2(\boldsymbol{S} \cdot \boldsymbol{N}')\boldsymbol{N}' - \boldsymbol{S} = 2(\boldsymbol{S} \cdot (\boldsymbol{N} + \delta\boldsymbol{N}))(\boldsymbol{N} + \delta\boldsymbol{N}) - \boldsymbol{S} \\ &= \boldsymbol{V} + 2(\boldsymbol{S} \cdot \delta\boldsymbol{N})\boldsymbol{N} + 2(\boldsymbol{S} \cdot \boldsymbol{N})\delta\boldsymbol{N} + 2(\boldsymbol{S} \cdot \delta\boldsymbol{N})\delta\boldsymbol{N}, \end{aligned}$$

where $\delta\boldsymbol{N} \stackrel{\text{def}}{=} \boldsymbol{N}' - \boldsymbol{N} = \delta s\, d\boldsymbol{N}(\boldsymbol{t})$. Because $\boldsymbol{t}$ is tangent to the surface in P, ignoring second-order terms yields

$$\delta a = (\boldsymbol{V} - \boldsymbol{V}') \cdot \boldsymbol{t} = 2(\boldsymbol{S} \cdot \boldsymbol{N})(\delta\boldsymbol{N} \cdot \boldsymbol{t}) = 2(\boldsymbol{S} \cdot \boldsymbol{N})(\delta s)(\mathrm{II}(\boldsymbol{t}, \boldsymbol{t})).$$

Thus, for a concave surface, the specularity always moves against the view, and for a convex surface, it always moves with the view. Things are more complex with hyperbolic surfaces; the specularity may move with the view, against the view, or perpendicular to the view (when $\boldsymbol{t}$ is an asymptotic direction).

[1] Of course, there is a simpler way to distinguish (by sight) the concave bowl of a spoon from its convex back: it is just the side where your reflection is upside down. This property of concave mirrors was demonstrated to one of the authors by one of his friends at a dinner party, causing much consternation since the author in question was at the time bragging about differential geometry, and his friend, who does not pretend to know anything about mathematics, just looked at herself in the spoon.

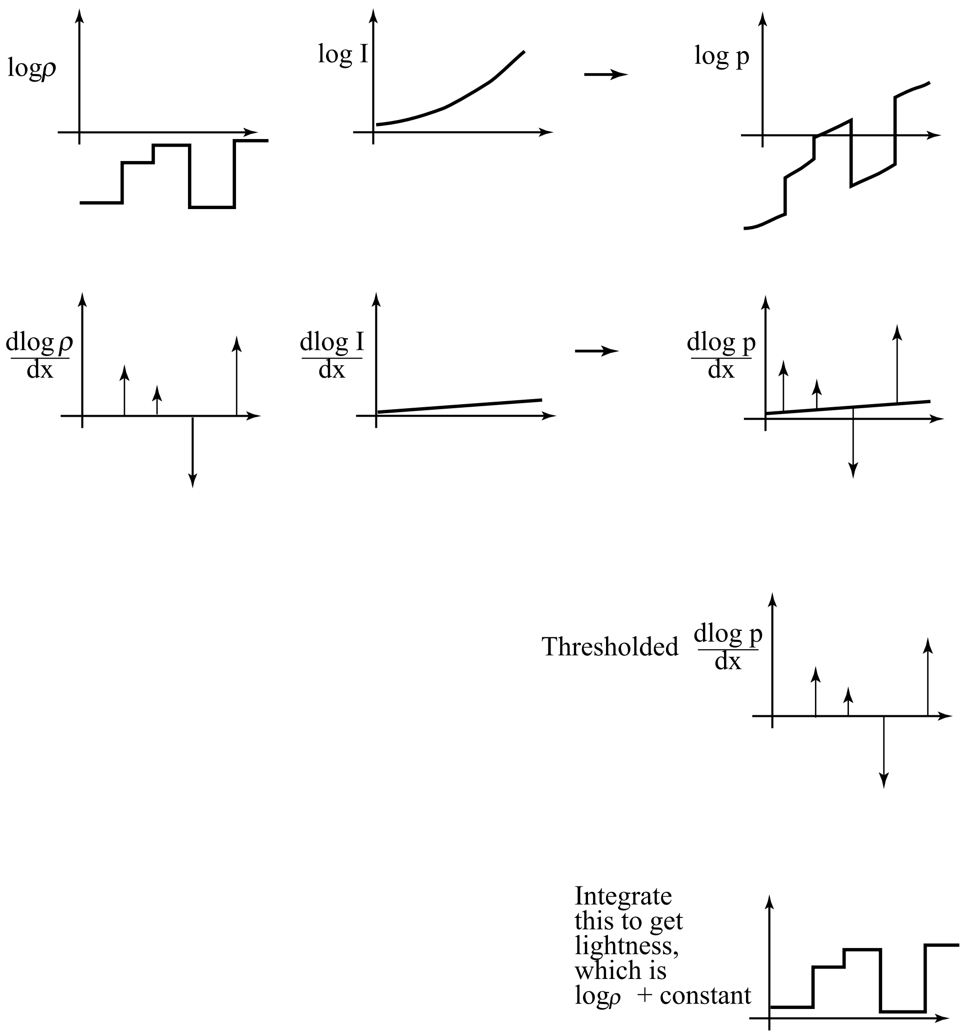

FIGURE 2.8: The lightness algorithm is easiest to illustrate for a 1D image. In the **top row**, the graph on the **left** shows $\log \rho(x)$, that in the **center** $\log I(x)$, and that on the **right** their sum, which is $\log C$. The log of image intensity has large derivatives at changes in surface reflectance and small derivatives when the only change is due to illumination gradients. Lightness is recovered by differentiating the log intensity, thresholding to dispose of small derivatives, and integrating at the cost of a missing constant of integration.

2.2.3 Inferring Lightness and Illumination

If we could estimate the albedo of a surface from an image, then we would know a property of the surface itself, rather than a property of a picture of the surface. Such properties are often called *intrinsic representations*. They are worth estimating, because they do not change when the imaging circumstances change. It might seem that albedo is difficult to estimate, because there is an ambiguity linking albedo and illumination; for example, a high albedo viewed under middling illumination will give the same brightness as a low albedo viewed under bright light. However, humans can report whether a surface is white, gray, or black (the *lightness* of the surface), despite changes in the intensity of illumination (the *brightness*). This skill is known as *lightness constancy.* There is a lot of evidence that human lightness constancy involves two processes: one process compares the brightness of various image patches and uses this comparison to determine which patches are lighter and which darker; the second establishes some form of absolute standard to which these comparisons can be referred (e.g. Gilchrist *et al.* (1999)).

Current lightness algorithms were developed in the context of simple scenes. In particular, we assume that the scene is flat and frontal; that surfaces are diffuse, or that specularities have been removed; and that the camera responds linearly. In this case, the camera response C at a point $\boldsymbol{x}$ is the product of an illumination term, an albedo term, and a constant that comes from the camera gain:

$$C(\boldsymbol{x}) = k_c I(\boldsymbol{x})\rho(\boldsymbol{x}).$$

If we take logarithms, we get

$$\log C(\boldsymbol{x}) = \log k_c + \log I(\boldsymbol{x}) + \log \rho(\boldsymbol{x}).$$

We now make a second set of assumptions:

- First, we assume that albedoes change only quickly over space. This means that a typical set of albedoes will look like a collage of papers of different grays. This assumption is quite easily justified: There are relatively few continuous changes of albedo in the world (the best example occurs in ripening fruit), and changes of albedo often occur when one object occludes another (so we would expect the change to be fast). This means that spatial derivatives of the term $\log \rho(\boldsymbol{x})$ are either zero (where the albedo is constant) or large (at a change of albedo).

- Second, illumination changes only slowly over space. This assumption is somewhat realistic. For example, the illumination due to a point source will change relatively slowly unless the source is very close, so the sun is a particularly good source for this method, as long as there are no shadows. As another example, illumination inside rooms tends to change very slowly because the white walls of the room act as area sources. This assumption fails dramatically at shadow boundaries, however. We have to see these as a special case and assume that either there are no shadow boundaries or that we know where they are.

We can now build algorithms that use our model. The earliest algorithm is the Retinex algorithm of Land and McCann (1971); this took several forms, most of which have fallen into disuse. The key insight of Retinex is that small gradients are changes in illumination, and large gradients are changes in lightness. We can use this by differentiating the log transform, throwing away small gradients, and integrating the results (Horn 1974); these days, this procedure is widely known as

Retinex. There is a constant of integration missing, so lightness ratios are available, but absolute lightness measurements are not. Figure 2.8 illustrates the process for a one-dimensional example, where differentiation and integration are easy.

This approach can be extended to two dimensions as well. Differentiating and thresholding is easy: at each point, we estimate the magnitude of the gradient; if the magnitude is less than some threshold, we set the gradient vector to zero; otherwise, we leave it alone. The difficulty is in integrating these gradients to get the log albedo map. The thresholded gradients may not be the gradients of an image because the mixed second partials may not be equal (integrability again; compare with Section 2.2.4).

Form the gradient of the log of the image
At each pixel, if the gradient magnitude is below
  a threshold, replace that gradient with zero
Reconstruct the log-albedo by solving the minimization
  problem described in the text
Obtain a constant of integration
Add the constant to the log-albedo, and exponentiate

Algorithm 2.1: Determining the Lightness of Image Patches.

The problem can be rephrased as a minimization problem: choose the log albedo map whose gradient is most like the thresholded gradient. This is a relatively simple problem because computing the gradient of an image is a linear operation. The x-component of the thresholded gradient is scanned into a vector $\boldsymbol{p}$, and the y-component is scanned into a vector $\boldsymbol{q}$. We write the vector representing log-albedo as $\boldsymbol{l}$. Now the process of forming the x derivative is linear, and so there is some matrix $\mathcal{M}_x$, such that $\mathcal{M}_x\boldsymbol{l}$ is the x derivative; for the y derivative, we write the corresponding matrix $\mathcal{M}_y$.

The problem becomes finding the vector $\boldsymbol{l}$ that minimizes

$$| \mathcal{M}_x\boldsymbol{l} - \boldsymbol{p} |^2 + | \mathcal{M}_y\boldsymbol{l} - \boldsymbol{q} |^2 .$$

This is a quadratic minimization problem, and the answer can be found by a linear process. Some special tricks are required because adding a constant vector to $\boldsymbol{l}$ cannot change the derivatives, so the problem does not have a unique solution. We explore the minimization problem in the exercises.

The constant of integration needs to be obtained from some other assumption. There are two obvious possibilities:

- we can assume that the *brightest patch is white*;
- we can assume that the *average lightness is constant.*

We explore the consequences of these models in the exercises.

More sophisticated algorithms are now available, but there were no quantitative studies of performance until recently. Grosse *et al.* built a dataset for

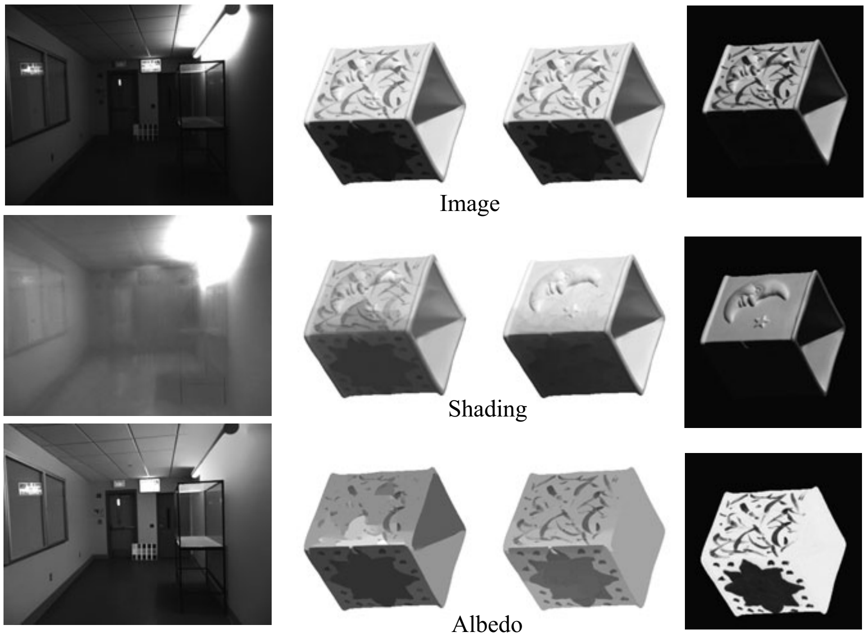

FIGURE 2.9: Retinex remains a strong algorithm for recovering albedo from images. Here we show results from the version of Retinex described in the text applied to an image of a room (**left**) and an image from a collection of test images due to Grosse *et al.* (2009). The **center-left** column shows results from Retinex for this image, and the **center-right** column shows results from a variant of the algorithm that uses color reasoning to improve the classification of edges into albedo versus shading. Finally, the **right** column shows the correct answer, known by clever experimental methods used when taking the pictures. This problem is very hard; you can see that the albedo images still contain some illumination signal. *Part of this figure courtesy Kevin Karsch, U. Illinois. Part of this figure was originally published as Figure 3 of "Ground truth dataset and baseline evaluations for intrinsic image algorithms," by R. Grosse, M. Johnson, E. Adelson, and W. Freeman, Proc. IEEE ICCV 2009, © IEEE, 2009.*

evaluating lightness algorithms, and show that a version of the procedure we describe performs extremely well compared to more sophisticated algorithms (2009). The major difficulty with all these approaches is caused by shadow boundaries, which we discuss in Section 3.5.2.

2.2.4 Photometric Stereo: Shape from Multiple Shaded Images

It is possible to reconstruct a patch of surface from a series of pictures of that surface taken under different illuminants. First, we need a camera model. For simplicity, we choose a camera situated so that the point (x, y, z) in space is imaged to the point (x, y) in the camera (the method we describe works for the other camera models described in Chapter 1).

In this case, to measure the shape of the surface, we need to obtain the depth to the surface. This suggests representing the surface as $(x, y, f(x, y))$—a representation known as a *Monge patch* after the French military engineer who first used it (Figure 2.10). This representation is attractive because we can determine a unique point on the surface by giving the image coordinates. Notice that to obtain a measurement of a solid object, we would need to reconstruct more than one patch because we need to observe the back of the object.

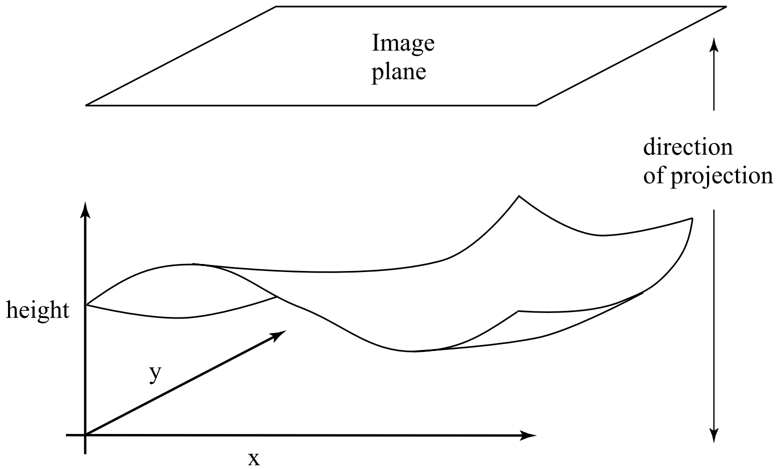

FIGURE 2.10: A Monge patch is a representation of a piece of surface as a height function. For the photometric stereo example, we assume that an orthographic camera—one that maps (x, y, z) in space to (x, y) in the camera—is viewing a Monge patch. This means that the shape of the surface can be represented as a function of position in the image.

Photometric stereo is a method for recovering a representation of the Monge patch from image data. The method involves reasoning about the image intensity values for several different images of a surface in a fixed view illuminated by different sources. This method recovers the height of the surface at points corresponding to each pixel; in computer vision circles, the resulting representation is often known as a *height map*, *depth map*, or *dense depth map*.

Fix the camera and the surface in position, and illuminate the surface using a point source that is far away compared with the size of the surface. We adopt a local shading model and assume that there is no ambient illumination (more about this later) so that the brightness at a point $\boldsymbol{x}$ on the surface is

$$B(\boldsymbol{x}) = \rho(\boldsymbol{x})\boldsymbol{N}(\boldsymbol{x}) \cdot \boldsymbol{S}_1,$$

where $\boldsymbol{N}$ is the unit surface normal and $\boldsymbol{S}_1$ is the source vector. We can write $B(x, y)$ for the radiosity of a point on the surface because there is only one point on the surface corresponding to the point (x, y) in the camera. Now we assume that the response of the camera is linear in the surface radiosity, and so have that the value of a pixel at (x, y) is

$$\begin{aligned} I(x,y) &= kB(\boldsymbol{x}) \\ &= kB(x,y) \\ &= k\rho(x,y)\boldsymbol{N}(x,y) \cdot \boldsymbol{S}_1 \\ &= \boldsymbol{g}(x,y) \cdot \boldsymbol{V}_1, \end{aligned}$$

where $\boldsymbol{g}(x,y) = \rho(x,y)\boldsymbol{N}(x,y)$ and $\boldsymbol{V}_1 = k\boldsymbol{S}_1$, where k is the constant connecting the camera response to the input radiance.

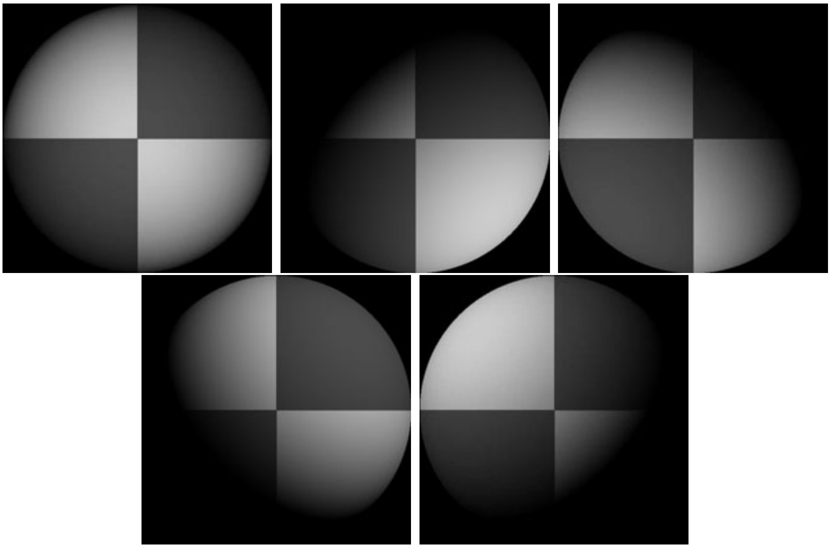

FIGURE 2.11: Five synthetic images of a sphere, all obtained in an orthographic view from the same viewing position. These images are shaded using a local shading model and a distant point source. This is a convex object, so the only view where there is no visible shadow occurs when the source direction is parallel to the viewing direction. The variations in brightness occuring under different sources code the shape of the surface.

In these equations, $\boldsymbol{g}(x,y)$ describes the surface, and $\boldsymbol{V}_1$ is a property of the illumination and of the camera. We have a dot product between a vector field $\boldsymbol{g}(x,y)$ and a vector $\boldsymbol{V}_1$, which could be measured; with enough of these dot products, we could reconstruct $\boldsymbol{g}$ and so the surface.

Now if we have n sources, for each of which $\boldsymbol{V}_i$ is known, we stack each of these $\boldsymbol{V}_i$ into a known matrix $\mathcal{V}$, where

$$\mathcal{V} = \begin{pmatrix} \boldsymbol{V}_1^T \\ \boldsymbol{V}_2^T \\ \dots \\ \boldsymbol{V}_n^T \end{pmatrix}.$$

For each image point, we stack the measurements into a vector

$$\boldsymbol{i}(x,y) = \{I_1(x,y), I_2(x,y), \dots, I_n(x,y)\}^T.$$

Notice that we have one vector per image point; each vector contains all the image brightnesses observed at that point for different sources. Now we have

$$\boldsymbol{i}(x, y) = \mathcal{V}\boldsymbol{g}(x, y),$$

and $\boldsymbol{g}$ is obtained by solving this linear system—or rather, one linear system per point in the image. Typically, $n > 3$, so that a least-squares solution is appropriate. This has the advantage that the residual error in the solution provides a check on our measurements.

Substantial regions of the surface might be in shadow for one or the other light (see Figure 2.11). We assume that all shadowed regions are known, and deal only with points that are not in shadow for any illuminant. More sophisticated strategies can infer shadowing because shadowed points are darker than the local geometry predicts.

We can extract the albedo from a measurement of $\boldsymbol{g}$ because $\boldsymbol{N}$ is the unit normal. This means that $|\boldsymbol{g}(x, y)| = \rho(x, y)$. This provides a check on our measurements as well. Because the albedo is in the range zero to one, any pixels where $|\boldsymbol{g}|$ is greater than one are suspect—either the pixel is not working or $\mathcal{V}$ is incorrect. Figure 2.12 shows albedo recovered using this method for the images shown in Figure 2.11.

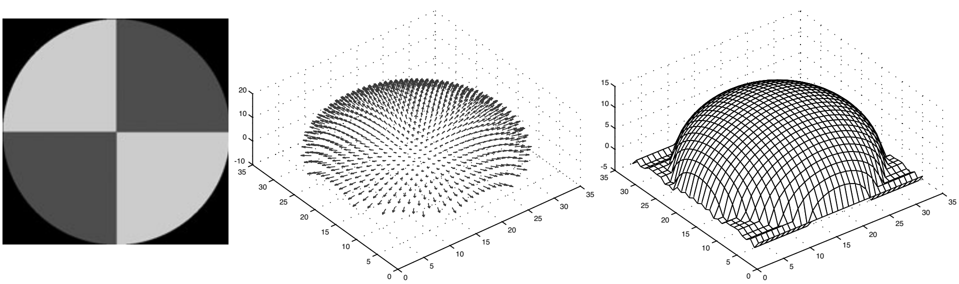

FIGURE 2.12: The image on the **left** shows the magnitude of the vector field $\boldsymbol{g}(x, y)$ recovered from the input data of Figure 2.11 represented as an image—this is the reflectance of the surface. The **center** figure shows the normal field, and the **right** figure shows the height field.

We can extract the surface normal from $\boldsymbol{g}$ because the normal is a unit vector

$$\boldsymbol{N}(x, y) = \frac{\boldsymbol{g}(x, y)}{|\boldsymbol{g}(x, y)|}.$$

Figure 2.12 shows normal values recovered for the images of Figure 2.11.

The surface is $(x, y, f(x, y))$, so the normal as a function of (x, y) is

$$\boldsymbol{N}(x, y) = \frac{1}{\sqrt{1 + \frac{\partial f}{\partial x}^2 + \frac{\partial f}{\partial y}^2}} \left\{ \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, 1 \right\}^T.$$

To recover the depth map, we need to determine $f(x, y)$ from measured values of the unit normal.

Obtain many images in a fixed view under different illuminants
Determine the matrix $\mathcal{V}$ from source and camera information

Inferring albedo and normal:
For each point in the image array that is not shadowed
  Stack image values into a vector $\boldsymbol{i}$
  Solve $\mathcal{V}\boldsymbol{g} = \boldsymbol{i}$ to obtain $\boldsymbol{g}$ for this point
  Albedo at this point is $|\boldsymbol{g}|$
  Normal at this point is $\frac{\boldsymbol{g}}{|\boldsymbol{g}|}$
  p at this point is $\frac{N_1}{N_3}$
  q at this point is $\frac{N_2}{N_3}$
end
Check: is $(\frac{\partial p}{\partial y} - \frac{\partial q}{\partial x})^2$ small everywhere?

Integration:
Top left corner of height map is zero
For each pixel in the left column of height map
  height value = previous height value + corresponding q value
end
For each row
  For each element of the row except for leftmost
    height value = previous height value + corresponding p value
  end
end

Algorithm 2.2: Photometric Stereo.

Assume that the measured value of the unit normal at some point (x, y) is $(a(x, y), b(x, y), c(x, y))$. Then

$$\frac{\partial f}{\partial x} = \frac{a(x,y)}{c(x,y)} \text{ and } \frac{\partial f}{\partial y} = \frac{b(x,y)}{c(x,y)}.$$

We have another check on our data set, because

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x},$$

so we expect that

$$\frac{\partial\left(\frac{a(x,y)}{c(x,y)}\right)}{\partial y} - \frac{\partial\left(\frac{b(x,y)}{c(x,y)}\right)}{\partial x}$$

should be small at each point. In principle it should be zero, but we would have to estimate these partial derivatives numerically and so should be willing to accept small values. This test is known as a test of *integrability*, which in vision applications always boils down to checking that mixed second partials are equal.

FIGURE 2.13: Photometric stereo could become the method of choice to capture complex deformable surfaces. On the **top**, three images of a garment, lit from different directions, which produce the reconstruction shown on the **top right**. A natural way to obtain three different images at the same time is to use a color camera; if one has a red light, a green light, and a blue light, then a single color image frame can be treated as three images under three separate lights. On the **bottom**, an image of the garment captured in this way, which results in the photometric stereo reconstruction on the **bottom right**. *This figure was originally published as Figure 6 of "Video Normals from Colored Lights," G. J. Brostow, C. Hernández, G. Vogiatzis, B. Stenger, and R. Cipolla, IEEE Transactions on Pattern Analysis and Machine Intelligence, 2011 © IEEE, 2011.*

Assuming that the partial derivatives pass this sanity test, we can reconstruct the surface up to some constant depth error. The partial derivative gives the change in surface height with a small step in either the x or the y direction. This means we can get the surface by summing these changes in height along some path. In particular, we have

$$f(x, y) = \oint_C \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \cdot d\boldsymbol{l} + c,$$

where C is a curve starting at some fixed point and ending at (x, y), and c is a constant of integration, which represents the (unknown) height of the surface at the start point. The recovered surface does not depend on the choice of curve (exercises). Another approach to recovering shape is to choose the function $f(x, y)$ whose partial derivatives most look like the measured partial derivatives. Figure 2.12 shows the reconstruction obtained for the data shown in Figure 2.11.

Current reconstruction work tends to emphasize geometric methods that reconstruct from multiple views. These methods are very important, but often require feature matching, as we shall see in Chapters 7 and 8. This tends to mean that it is hard to get very high spatial resolution, because some pixels are consumed in resolving features. Recall that resolution (which corresponds roughly to the spatial frequencies that can be reconstructed accurately) is not the same as accuracy (which involves a method providing the right answers for the properties it estimates). Feature-based methods are capable of spectacularly accurate reconstructions. Because photometric cues have such spatial high resolution, they are a topic of considerable current interest. One way to use photometric cues is to try and match pixels with the same brightness across different cameras; this is difficult, but produces impressive reconstructions. Another is to use photometric stereo ideas. For some applications, photometric stereo is particularly atractive because one can get reconstructions from a single view direction—this is important, because we cannot always set up multiple cameras. In fact, with a trick, it is possible to get reconstructions from a single frame. A natural way to obtain three different images at the same time is to use a color camera; if one has a red light, a green light and a blue light, then a single color image frame can be treated as three images under three separate lights, and photometric stereo methods apply. In turn, this means that photometric stereo methods could be used to recover high-resolution reconstructions of deforming surfaces in a relatively straightforward way. This is particularly useful when it is difficult to get many cameras to view the object. Figure 2.13 shows one application to reconstructing cloth in video (from Brostow *et al.* (2011)), where multiple view reconstruction is complicated by the need to synchronize frames (alternatives are explored in, for example, White *et al.* (2007) or Bradley *et al.* (2008*b*)).

2.3 MODELLING INTERREFLECTION

The difficulty with a local shading model is that it doesn't account for all light. The alternative is a *global shading model*, where we account for light arriving from other surfaces as well as from the luminaire. As we shall see, such models are tricky to work with. In such models, each surface patch receives power from all the radiating surfaces it can see. These surfaces might radiate power that they generate internally because they are luminaires, or they might simply reflect power. The general form of the model will be:

$$\left(\begin{array}{c}\text{Power leaving}\\ \text{a patch}\end{array}\right) = \left(\begin{array}{c}\text{Power generated}\\ \text{by that patch}\end{array}\right) + \left(\begin{array}{c}\text{Power received from}\\ \text{other patches and reflected}\end{array}\right)$$

This means we need to be able to model the power received from other patches and reflected. We will develop a model assuming that all surfaces are diffuse. This leads to a somewhat simpler model, and describes all effects that are currently of interest to vision (it is complicated, but not difficult, to build more elaborate models). We will also need some radiometric terminology.

2.3.1 The Illumination at a Patch Due to an Area Source

The appropriate unit for illumination is **radiance**, defined as

> the power (amount of energy per unit time) traveling at some point in a specified direction, per unit area *perpendicular to the direction of travel*, per unit solid angle.

The units of radiance are watts per square meter per steradian ($Wm^{-2}sr^{-1}$). The definition of radiance might look strange, but it is consistent with the most basic phenomenon in radiometry: the amount of energy a patch collects from a source depends both on how large the source looks from the patch *and* on how large the patch looks from the source.

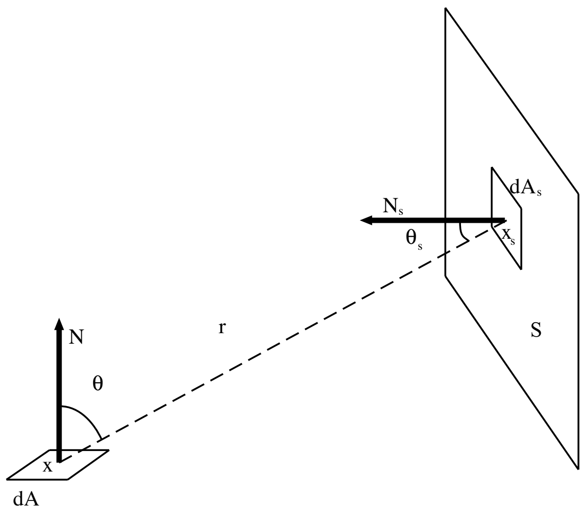

FIGURE 2.14: A patch with area dA views an area source S. We compute the power received by the patch by summing the contributions of each element on S, using the notation indicated in this figure.

It is important to remember that the square meters in the units for radiance are *foreshortened* (i.e., perpendicular to the direction of travel) to account for this phenomenon. Assume we have two elements, one at $\boldsymbol{x}$ with area dA and the other at $\boldsymbol{x}_s$ with area dA_s. Write the angular direction from $\boldsymbol{x}$ to $\boldsymbol{x}_s$ as $\boldsymbol{x} \to \boldsymbol{x}_s$, and define the angles θ and θ_s as in Figure 2.14. Then the solid angle subtended by element 2 at element 1 is

$$d\omega_{2(1)} = \frac{\cos\theta_s dA_s}{r^2},$$

so the power leaving $\boldsymbol{x}$ toward $\boldsymbol{x}_s$ is

$$\begin{aligned} d^2P_{1\to 2} &= \text{(radiance)(foreshortened area)(solid angle)} \\ &= L(\boldsymbol{x}, \boldsymbol{x} \to \boldsymbol{x}_s)(\cos\theta dA)(d\omega_{2(1)}) \\ &= L(\boldsymbol{x}, \boldsymbol{x} \to \boldsymbol{x}_s)\left(\frac{\cos\theta\cos\theta_s}{r^2}\right) dA_s dA. \end{aligned}$$

By a similar argument, the same expression yields the power arriving at $\boldsymbol{x}$ from $\boldsymbol{x}_2$; this means that, in a vacuum, *radiance is constant along (unoccluded) straight lines.*

We can now compute the power that an element dA collects from an area source, by summing the contributions of elements over that source. Using the notation of Figure 2.14, we get

$$dP_{S\to dA} = \left(\int_S L(\boldsymbol{x}_s, \boldsymbol{x}_s \to \boldsymbol{x})\left(\frac{\cos\theta_s\cos\theta}{r^2}\right) dA_s\right) dA.$$

To get a more useful area source model, we need further units.

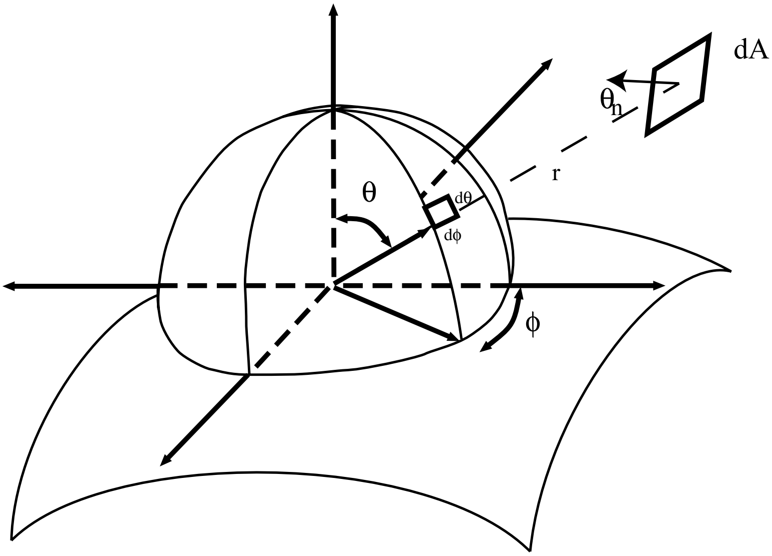

FIGURE 2.15: A hemisphere on a patch of surface, to show our angular coordinates for computing radiometric quantities. The coordinate axes are there to help you see the drawing as a 3D surface. An infinitesimal patch of surface with area dA which is distance r away is projected onto the unit hemisphere centered at the relevant point; the resulting area is the solid angle of the patch, marked as $d\theta d\phi$. In this case, the patch is small so that the area and hence the solid angle is $(1/r^2)dA\cos\theta_n$, where θ_n is the angle of inclination of the patch.

2.3.2 Radiosity and Exitance

We are dealing with diffuse surfaces, and our definition of a diffuse surface is that the intensity (formally, the radiance) leaving the surface is independent of the direction in which it leaves. There is no point in describing the intensity of such a surface with radiance (which explicitly depends on direction). The appropriate unit is *radiosity*, defined as

> the total power leaving a point on a surface per unit area on the surface.

Radiosity, which is usually written as $B(\boldsymbol{x})$, has units watts per square meter (Wm^{-2}). To obtain the radiosity of a surface at a point, we can sum the radiance leaving the surface at that point over the whole exit hemisphere. Thus, if $\boldsymbol{x}$ is a point on a surface emitting radiance $L(\boldsymbol{x}, \theta, \phi)$, the radiosity at that point is

$$B(\boldsymbol{x}) = \int_{\Omega} L(\boldsymbol{x}, \theta, \phi)\cos\theta d\omega,$$

where Ω is the exit hemisphere, $d\omega$ is unit solid angle, and the term $\cos\theta$ turns foreshortened area into area (look at the definitions of radiance and of radiosity again). We could substitute $d\omega = \sin\theta d\theta d\phi$, using the units of Figure 2.15.

Consider a surface element as in Figure 2.14. We have computed how much power it receives from the source as a function of the source's radiance. The surface element is diffuse, and its albedo is $\rho(\boldsymbol{x})$. The albedo is the fraction of incoming power that the surface radiates, so the radiosity due to power received from the area source is

$$B(\boldsymbol{x}) = \frac{dP_{S\to dA}}{dA} = \rho(\boldsymbol{x}) \left(\int_S L(\boldsymbol{x}_s, \boldsymbol{x}_s \to \boldsymbol{x}) \left(\frac{\cos\theta_s \cos\theta}{r^2} \right) dA_s \right).$$

Now if a point $\boldsymbol{u}$ on a surface has radiosity $B(\boldsymbol{u})$, what is the radiance leaving the surface in some direction? We write L for the radiance, which is independent of angle, and we must have

$$B(\boldsymbol{u}) = \int_\Omega L(\boldsymbol{x}, \theta, \phi) \cos\theta d\omega = L(\boldsymbol{u}) \int_\Omega \cos\theta d\omega = L(\boldsymbol{u})\pi.$$

This means that if the area source has radiosity $B(\boldsymbol{x}_s)$, then the radiosity at the element *due to the power received from the area source* is

$$\begin{aligned} B(\boldsymbol{x}) &= \rho \left(\int_S L(\boldsymbol{x}_s, \boldsymbol{x}_s \to \boldsymbol{x}) \left(\frac{\cos\theta_s \cos\theta}{r^2} \right) dA_s \right) \\ &= \rho \left(\int_S \frac{B(\boldsymbol{x})}{\pi} \left(\frac{\cos\theta_s \cos\theta}{r^2} \right) dA_s \right) \\ &= \frac{\rho}{\pi} \left(\int_S B(\boldsymbol{x}) \left(\frac{\cos\theta_s \cos\theta}{r^2} \right) dA_s \right). \end{aligned}$$

Our final step is to model illumination generated internally in a surface—light generated by a luminaire, rather than reflected from a surface. We assume there are no directional effects in the luminaire and that power is uniformly distributed across outgoing directions (this is the least plausible component of the model, but is usually tolerable). We use the unit *exitance*, which is defined as

> the total power internally generated power leaving a point on a surface per unit area on the surface.

2.3.3 An Interreflection Model

We can now write a formal model of interreflections for diffuse surfaces by substituting terms into the original expression. Recall that radiosity is power per unit area, write $E(\boldsymbol{x})$ for exitance at the point $\boldsymbol{x}$, write $\boldsymbol{x}_s$ for a coordinate that runs over all surface patches, $\mathcal{S}$ for the set of all surfaces, dA for the element of area at $\boldsymbol{x}$, $\mathrm{V}(\boldsymbol{x}, \boldsymbol{x}_s)$ for a function that is one if the two points can see each other and zero otherwise, and $\cos\theta$, $\cos\theta_s$, r, as in Figure 2.14. We obtain

$$\begin{array}{cccc} \text{Power leaving} & = & \text{Power generated} & + & \text{Power received from} \\ \text{a patch} & & \text{by that patch} & & \text{other patches and reflected} \\ & & & & \\ B(\boldsymbol{x})dA & = & E(\boldsymbol{x})dA & + & \frac{\rho(\boldsymbol{x})}{\pi} \int_{\mathcal{S}} \left[\begin{array}{c} \frac{\cos\theta \cos\theta_s}{r^2} \\ \times \\ \mathrm{V}(\boldsymbol{x}, \boldsymbol{x}_s) \end{array} \right] B(\boldsymbol{x_s}) dA_s dA \end{array}$$

and so, dividing by area, we have

$$B(\boldsymbol{x}) \quad = \quad E(\boldsymbol{x}) \quad + \quad \frac{\rho(\boldsymbol{x})}{\pi} \int_{\mathcal{S}} \left[\frac{\cos\theta\cos\theta_s}{r^2} \mathrm{Vis}(\boldsymbol{x}, \boldsymbol{x}_s)\right] B(\boldsymbol{x_s}) dA_s.$$

It is usual to write

$$K(\boldsymbol{x}, \boldsymbol{x}_s) = \frac{\cos\theta\cos\theta_s}{\pi r^2}$$

and refer to K as the interreflection kernel. Substituting gives

$$B(\boldsymbol{x}) = E(\boldsymbol{x}) + \rho(\boldsymbol{x}) \int_{\mathcal{S}} K(\boldsymbol{x}, \boldsymbol{x}_s) \mathrm{Vis}(\boldsymbol{x}, \boldsymbol{x}_s) B(\boldsymbol{x}_s) dA_{\boldsymbol{x}_s}$$

an equation where the solution appears inside the integral. Equations of this form are known as Fredholm integral equations of the second kind. This particular equation is a fairly nasty sample of the type because the interreflection kernel generally is not continuous and may have singularities. Solutions of this equation can yield quite good models of the appearance of diffuse surfaces, and the topic supports a substantial industry in the computer graphics community (good places to start for this topic are Cohen and Wallace (1993) or Sillion (1994)). The model produces good predictions of observed effects (Figure 2.16).

2.3.4 Qualitative Properties of Interreflections

Interreflections are a problem, because they are difficult to account for in our illumination model. For example, photometric stereo as we described it uses the model that light at a surface patch comes only from a distant light source. One could refine the method to take into account nearby light sources, but it is much more difficult to deal with interreflections. Once one accounts for interreflections, the brightness of each surface patch could be affected by the configuration of every other surface patch, making a very nasty global inference problem. While there have been attempts to build methods that can infer shape in the presence of interreflections (Nayar *et al.* 1991*a*), the problem is extremely difficult. One source of difficulties is that one may need to account for every radiating surface in the solution, even distant surfaces one cannot see.

An alternative strategy to straightforward physical inference is to understand the qualitative properties of interreflected shading. By doing so, we may be able to identify cases that are easy to handle, the main types of effect, and so on. The effects can be quite large. For example, Figure 2.17 shows views of the interior of two rooms. One room has black walls and contains black objects. The other has white walls and contains white objects. Each is illuminated (approximately!) by a distant point source. Given that the intensity of the source is adjusted appropriately, the local shading model predicts that these pictures would be indistinguishable. In fact, the black room has much darker shadows and crisper boundaries at the creases of the polyhedra than the white room. This is because surfaces in the black room reflect less light onto other surfaces (they are darker), whereas in the white room other surfaces are significant sources of radiation. The sections of the camera response to the radiosity (these are proportional to radiosity for diffuse surfaces) shown in the figure are hugely different qualitatively. In the black room, the radiosity is constant in patches, as a local shading model would predict, whereas in the white room slow

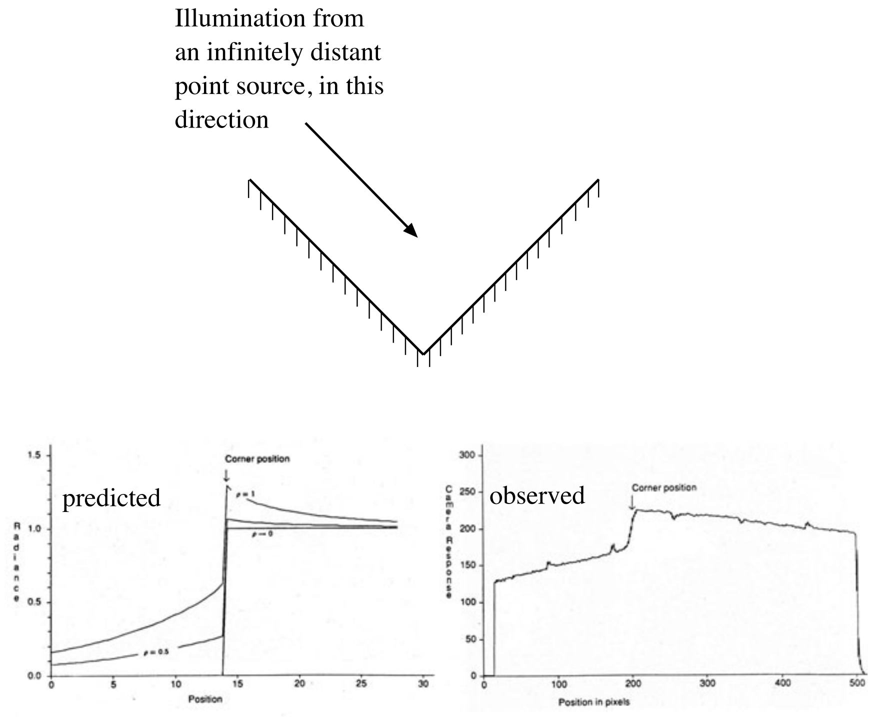

FIGURE 2.16: The model described in the text produces quite accurate qualitative predictions for interreflections. The **top** figure shows a concave right-angled groove illuminated by a point source at infinity where the source direction is parallel to the one face. On the **left** of the bottom row is a series of predictions of the radiosity for this configuration. These predictions have been scaled to lie on top of one another; the case $\rho \to 0$ corresponds to the local shading model. On the **right**, an observed image intensity for an image of this form for a corner made of white paper, showing the roof-like gradient in radiosity associated with the edge. A local shading model predicts a step. *This figure was originally published as Figures 5 and 7 of "Mutual Illumination," by D.A. Forsyth and A.P. Zisserman, Proc. IEEE CVPR, 1989, © IEEE, 1989.*

image gradients are quite common; these occur in concave corners, where object faces reflect light onto one another.

First, interreflections have a characteristic smoothing effect. This is most obviously seen when one tries to interpret a stained glass window by looking at the pattern it casts on the floor; this pattern is almost always a set of indistinct colored blobs. The effect is seen most easily with the crude model illustrated in Figure 2.18. The geometry consists of a patch with a frontal view of an infinite plane, which is a unit distance away and carries a radiosity $\sin \omega x$. There is no reason to vary the distance of the patch from the plane, because interreflection problems have scale invariant solutions, which means that the solution for a patch two units away can be obtained by reading our graph at 2ω. The patch is small enough that its contribution to the plane's radiosity can be ignored. If the patch is

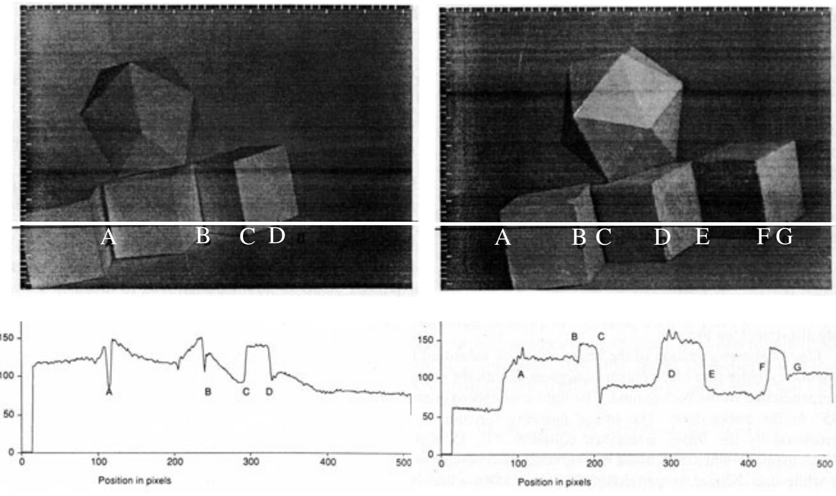

FIGURE 2.17: The column on the **left** shows data from a room with matte black walls and containing a collection of matte black polyhedral objects; that on the **right** shows data from a white room containing white objects. The images are qualitatively different, with darker shadows and crisper boundaries in the black room and bright reflexes in the concave corners in the white room. The graphs show sections of the image intensity along the corresponding lines in the images. *This figure was originally published as Figures 17, 18, 19, and 20 of "Mutual Illumination," by D.A. Forsyth and A.P. Zisserman, Proc. IEEE CVPR, 1989, © IEEE, 1989.*

slanted by σ with respect to the plane, it carries radiosity that is nearly periodic, with spatial frequency $\omega \cos \sigma$. We refer to the amplitude of the component at this frequency as the gain of the patch and plot the gain in Figure 2.18. The important property of this graph is that high spatial frequencies have a difficult time jumping the gap from the plane to the patch. This means that shading effects with high spatial frequency and high amplitude generally cannot come from distant surfaces (unless they are abnormally bright).

The extremely fast fall-off in amplitude with spatial frequency of terms due to distant surfaces means that, if one observes a high-amplitude term at a high spatial frequency, *it is very unlikely to have resulted from the effects of distant, passive radiators* (because these effects die away quickly). There is a convention, which we see in Section 2.2.3, that classifies effects in shading as due to reflectance if they are fast ("edges") and the dynamic range is relatively low and due to illumination otherwise. We can expand this convention. There is a mid range of spatial frequencies that are largely unaffected by mutual illumination from distant surfaces because the gain is small. Spatial frequencies in this range cannot be transmitted by distant passive radiators unless these radiators have improbably high radiosity. As a result, spatial frequencies in this range can be thought of as *regional properties*, which can result only from interreflection effects within a region.

The most notable regional properties are probably *reflexes*— small bright patches that appear mainly in concave regions (illustrated in Figure 2.19). A second

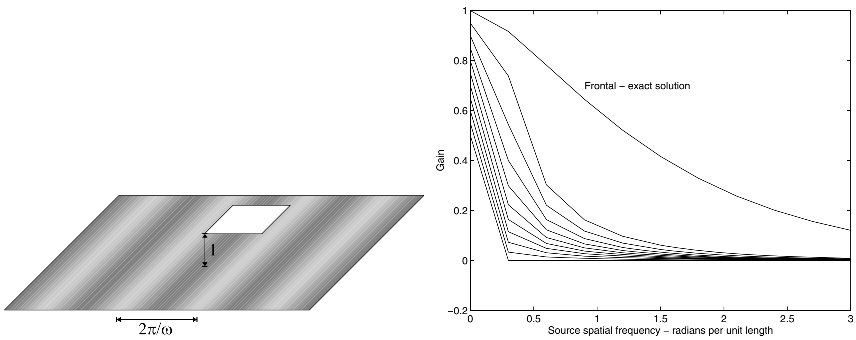

FIGURE 2.18: A small patch views a plane with sinusoidal radiosity of unit amplitude. This patch has a (roughly) sinusoidal radiosity due to the effects of the plane. We refer to the amplitude of this component as the *gain of the patch*. The graph shows numerical estimates of the gain for patches at 10 equal steps in slant angle, from 0 to $\pi/2$, as a function of spatial frequency *on the plane*. The gain falls extremely fast, meaning that large terms at high spatial frequencies must be regional effects, rather than the result of distant radiators. This is why it is hard to determine the pattern in a stained glass window by looking at the floor at the foot of the window. *This figure was originally published as Figures 1 and 2 from "Shading Primitives: Finding Folds and Shallow Grooves," J. Haddon and D.A. Forsyth, Proc. IEEE ICCV, 1998 © IEEE, 1998.*

important effect is *color bleeding*, where a colored surface reflects light onto another colored surface. This is a common effect that people tend not to notice unless they are consciously looking for it. It is quite often reproduced by painters.

2.4 SHAPE FROM ONE SHADED IMAGE

There is good evidence that people get some perception of shape from the shading pattern in a single image, though the details are uncertain and quite complicated (see the notes for a brief summary). You can see this evidence in practice: whenever you display a reconstruction of a surface obtained from images, it is a good idea to shade that reconstruction using image pixels, because it always looks more accurate. In fact, quite bad reconstructions can be made to look good with this method. White and Forsyth (2006) use this trick to replace surface albedos in movies; for example, they can change the pattern on a plastic bag in a movie. Their method builds and tracks very coarse geometric reconstructions, uses a form of regression to recover the original shading pattern of the object, and then shades the coarse geometric reconstruction using the original shading pattern (Figure 2.20). In this figure, the pictures look plausible, not because the reconstruction is good (it isn't), but because the shading pattern masks the errors in geometric reconstruction.

The cue to shape must come from the fact that a surface patch that faces the light source is brighter than one that faces away from the source. But going from this observation to a working algorithm remains an open question. The key seems to be an appropriate use of the *image irradiance equation*. Assume we have a surface in the form $(x, y, f(x, y))$ viewed orthographically along the z-axis. Assume that the surface is diffuse, and its albedo is uniform and known. Assume also that

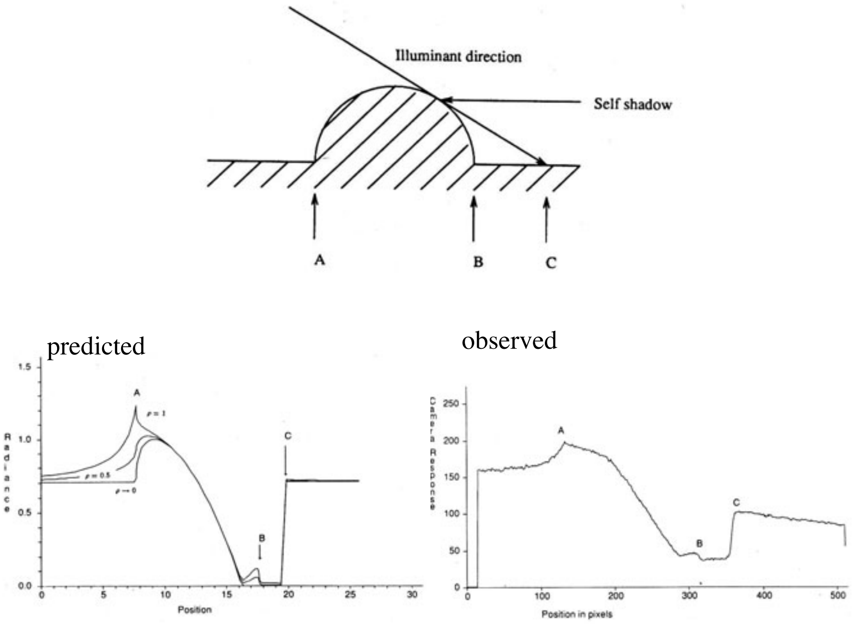

FIGURE 2.19: Reflexes occur quite widely; they are usually caused by a favorable view of a large reflecting surface. In the geometry shown on the **top**, the shadowed region of the cylindrical bump sees the plane background at a fairly favorable angle: if the background is large enough, nearly half the hemisphere of the patch at the base of the bump is a view of the plane. This means there will be a reflex with a large value attached to the edge of the bump and inside the cast shadow region (which a local model predicts as black). There is another reflex on the other side, too, as the series of solutions (again normalized for easy comparison) on the **left** show. On the **right**, an observation of this effect in a real scene. *This figure was originally published as Figures 24 and 26 of "Mutual Illumination," by D.A. Forsyth and A.P. Zisserman, Proc. IEEE CVPR, 1989, © IEEE, 1989.*

the model of Section 2.1.3 applies, so that the shading at a point with normal $\boldsymbol{N}$ is given by some function $R(\boldsymbol{N})$ (the function of our model is $R(\boldsymbol{N}) = \boldsymbol{N} \cdot \boldsymbol{S}$, but others could be used). Now the normal of our surface is a function of the two first partial derivatives

$$p = \frac{\partial f}{\partial x}, \; q = \frac{\partial f}{\partial y}$$

so we can write $R(p, q)$. Assume that the camera is radiometrically calibrated, so we can proceed from image values to intensity values. Write the intensity at x, y as $I(x, y)$. Then we have

$$R(p, q) = I(x, y).$$

This is a first order partial differential equation, because p and q are partial derivatives of f. In principle, we could set up some boundary conditions and solve this equation. Doing so reliably and accurately for general images remains outside our competence, 40 years after the problem was originally posed by Horn (1970a).

FIGURE 2.20: On the **left**, an original frame from a movie sequence of a deforming plastic bag. On the **right**, two frames where the original texture has been replaced by another. The method used is a form of regression; its crucial property is that it has a very weak geometric model, but is capable of preserving the original shading field of the image. If you look closely at the *albedo* (i.e., the black pattern) of the bag, you may notice that it is inconsistent with the wrinkles on the bag, but because the shading has been preserved, the figures look quite good. This is indirect evidence that shading is a valuable cue to humans. Little is known about how this cue is to be exploited, however. *This figure was originally published as Figure 10 of "Retexturing single views using texture and shading," by R. White and D.A. Forsyth, Proc. European Conference on Computer Vision. Springer Lecture Notes in Computer Science, Volume 3954, 2006 © Springer 2006.*

There are a variety of difficulties here. The physical model is a poor model of what actually happens at surfaces because any particular patch is illuminated by other surface patches, as well as by the source. We expect to see a rich variety of geometric constraints on the surface we reconstruct, and it is quite difficult to formulate shape from shading in a way that accomodates these constraints and still has a solution. Shading is a worthwhile cue to exploit, because we can observe shading at extremely high spatial resolutions, but this means we must work with very high dimensional models to reconstruct. Some schemes for shading reconstruction can be unstable, but there appears to be no theory to guide us to stable schemes. We very seldom actually see isolated surfaces of known albedo, and there are no methods that are competent to infer both shading and albedo, though there is some reason to hope that such methods can be built. We have no theory that is capable of predicting the errors in shading-based reconstructions from first principles. All this makes shape inference from shading in a single image one of the most frustrating open questions in computer vision.

2.5 NOTES

Horn started the systematic study of shading in computer vision, with important papers on recovering shape from a local shading model using a point source (in (Horn 1970*b*), (Horn 1975)), with a more recent account in Horn (1990).

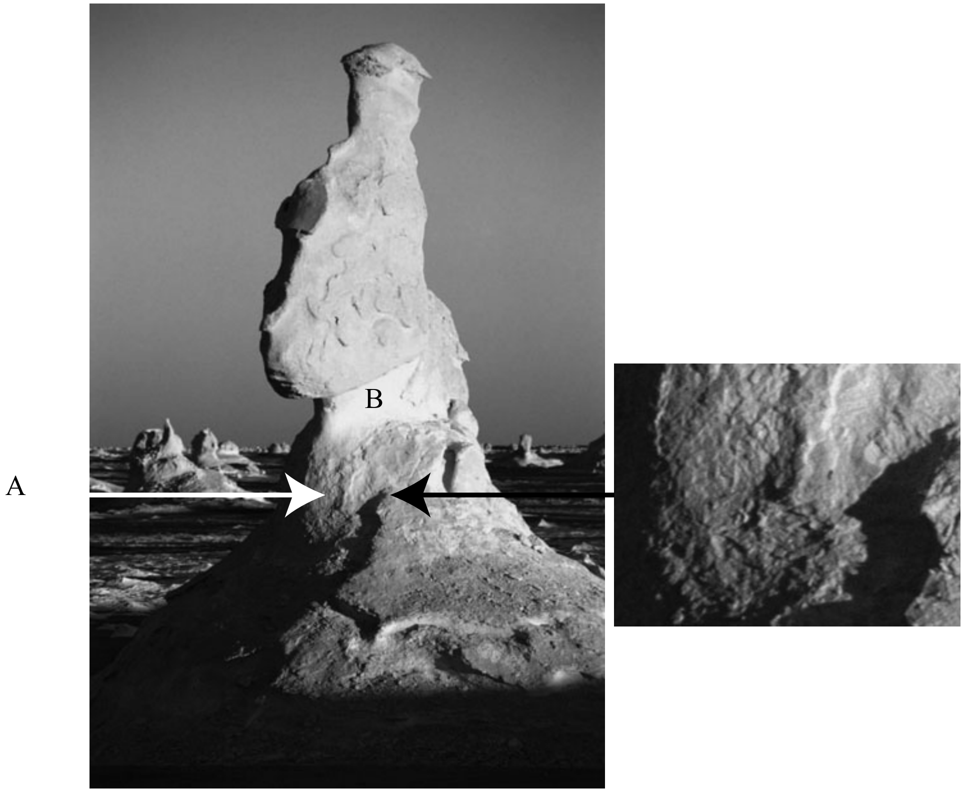

FIGURE 2.21: This picture shows two important mechanisms by which it might be possible to infer surface shape from single images. First, patches that face away from the light (like A, on the **left**) are darker than those that face the light (B). Second, shadows pick out relief—for example, small dents in a surface (more easily seen in the detail patch on the **right**), have a bright face facing the light and a dark face which is in shadow. *Peter Wilson © Dorling Kindersley, used with permission.*

Models of Shading

The first edition of this book contained more formal radiometry, which was widely disliked (and for good reason; making the subject exciting is beyond our skills). We've cut this down, and tried to avoid using the ideas, but point those who really want to know more toward that earlier edition. We strongly recommend François Sillion's excellent book (Sillion 1994) for its clear account of radiometric calculations. There are a variety of more detailed publications for reference (Nayar *et al.* 1991*c*). Our discussion of reflection is thoroughly superficial. The specular plus diffuse model appears to be originally due to Cook, Torrance, and Sparrow (Torrance and Sparrow 1967, Cook and Torrance 1987). A variety of modifications of this model appear in computer vision and computer graphics. Reflection models can be derived by combining a statistical description of surface roughness with electromagnetic considerations (e.g., Beckmann and Spizzichino (1987)) or by adopting scattering models (as in the work of Torrance and Sparrow (1967) and of Cook and Torrance (1987)).

It is commonly believed that rough surfaces are Lambertian. This belief has a substantial component of wishful thinking because rough surfaces often have local shadowing effects that make the radiance reflected quite strongly dependent on the illumination angle. For example, a stucco wall illuminated at a near grazing angle shows a clear pattern of light and dark regions where facets of the surface face toward the light or are shadowed. If the same wall is illuminated along the normal, this pattern largely disappears. Similar effects at a finer scale are averaged to endow rough surfaces with measurable departures from a Lambertian model (for details, see Koenderink *et al.* (1999), Nayar and Oren (1993), (1995), Oren and Nayar (1995), and Wolff *et al.* (1998)).

Inference from Shading

Registered images are not essential for radiometric calibration. For example, it is sufficient to have two images where we believe the histogram of E_{ij} values is the same (Grossberg and Nayar 2002). This occurs, for example, when the images are of the same scene, but are not precisely registered. Patterns of intensity around edges also can reveal calibration (Lin *et al.* 2004).

There has not been much recent study of lightness constancy algorithms. The basic idea is due to Land and McCann (1971). Their work was formalized for the computer vision community by Horn (1974). A variation on Horn's algorithm was constructed by Blake (1985). This is the lightness algorithm we describe. It appeared originally in a slightly different form, where it was called the *Retinex* algorithm (Land and McCann 1971). Retinex was originally intended as a color constancy algorithm. It is surprisingly difficult to analyze (Brainard and Wandell 1986).

Retinex estimates the log-illumination term by subtracting the log-albedo from the log-intensity. This has the disadvantage that we do not impose any structural constraints on illumination. This point has largely been ignored, because the main focus has been on albedo estimates. However, albedo estimates are likely to be improved by balancing violations of albedo constraints with those of illumination constraints.

Lightness techniques are not as widely used as they should be, particularly given that there is some evidence that they produce useful information on real images (Brelstaff and Blake 1987). Classifying illumination versus albedo simply by looking at the magnitude of the gradient is crude, and ignores important cues. Sharp shading changes occur at shadow boundaries or normal discontinuities, but using chromaticity (Funt *et al.* 1992) or multiple images under different lighting conditions (Weiss 2001) yields improved estimates. One can learn to distinguish illumination from albedo (Freeman *et al.* 2000). Discriminative methods to classify edges into albedo or shading help (Tappen *et al.* 2006*b*) and chromaticity cues can contribute (Farenzena and Fusiello 2007). Shading and albedo are sometimes known as *intrinsic images*. Tappen *et al.* (2006*a*) regress local intrinsic image patches against the image, exploiting the constraint that patches join up. When more than one image is available, recent methods can recover quite complex surface properties (Romeiro *et al.* 2008). When geometry is available, Yu *et al.* (1999) showed significant improvements in lightness recovery are possible.

In its original form, photometric stereo is due to Woodham. There are a number of variants of this useful idea (Horn *et al.* (1978), Woodham (1979), (1980), (1989), (1994), Woodham *et al.* (1991)). Current methods for photometric stereo require at least two unshadowed views; see Hernandez *et al.* (2008) which describes methods to cope in this case. There are a variety of variations on photometric stereo. Color photometric stereo seems to date to Petrov (1987), with a variant in Petrov (1991).

Photometric stereo depends only on adopting a local shading model. This model need not be a Lambertian surface illuminated by a distant point source. If the brightness of the surface is a known function of the surface normal satisfying a small number of constraints, photometric stereo is still possible. This is because the intensity of a pixel in a single view determines the normal up to a one-parameter family. This means that two views determine the normal. The simplest example of this case occurs for a surface of known albedo illuminated by a distant point source.

In fact, if the radiosity of the surface is a k-parameter function of the surface normal, photometric stereo is still possible. The intensity of the pixel in a single view determines the normal up to a $k+1$ parameter family, and $k+1$ views give the normal. For this approach to work, the brightness needs to be given by a function for which our arithmetic works (e.g., if the brightness of the surface is a constant function of the surface normal, it isn't possible to infer any constraint on the normal from the brightness). One can then recover shape and reflectance maps simultaneously (Garcia-Bermejo *et al.* (1996); Mukawa (1990); Nayar *et al.* (1990); Tagare and de Figueiredo (1992); (1993)).

A converse to photometric stereo might be as follows: Assume we have a diffuse sphere, immersed in an environment where illumination depends only on direction. What can we determine about the illumination field from the surface brightness? The answer is very little, because diffuse surfaces engage in a form of averaging that heavily smoothes the illumination field (Ramamoorthi and Hanrahan 2001). This is valuable because it suggests that complex representations of the directional properties illumination aren't required in a diffuse world. For example, this result allowed Jacobs (1981) to produce a form of photometric stereo that requires no illuminant information, using sufficient images.

Interreflections

The effects of global shading are often ignored in the shading literature, which causes a reflex response of hostility in one of the authors. The reason to ignore interreflections is that they are extremely hard to analyze, particularly from the perspective of inferring object properties given the output of a global shading model. If interreflection effects do not change the output of a method much, then it is probably all right to ignore them. Unfortunately, this line of reasoning is seldom pursued because it is quite difficult to show that a method is stable under interreflections. The discussion of spatial frequency issues follows Haddon and Forsyth (1998*a*), after an idea of Koenderink and van Doorn (1983). Apart from this, there is not much knowledge about the overall properties of interreflected shading, which is an important gap in our knowledge. An alternative strategy is to iteratively reestimate shape using a rendering model (Nayar *et al.* 1991*b*).

Horn is also the first author to indicate the significance of global shading effects (Horn 1977). Koenderink and van Doorn (1983) noted that the radiosity under a global model is obtained by taking the radiosity under a local model, and applying a linear operator. One then studies that operator; in some cases, its eigenfunctions (often called *geometrical modes*) are informative. Forsyth and Zisserman (1989, 1990, 1991) then demonstrated a variety of the qualitative effects due to interreflections.

Shape from One Shaded Image

Shape from shading is an important puzzle. Comprehensive surveys include (Horn and Brooks (1989); Zhang *et al.* (1999); Durou *et al.* (2008*b*)). In practice, despite the ongoing demand for high-resolution shape reconstructions, shape-from-shading has been a disappointment. This may be because, as currently formulated, it solves a problem that doesn't commonly occur. Image irradiance equation methods are formulated to produce reconstructions when there is very little geometric data, but it is much more common to want to improve the resolution of a method that already produces quite rich geometric data.

Methods are either too fragile or the reconstructions too poor for the method to be useful. Some of this may be due to the effects of interreflections. Another source of difficulty could be the compromises that need to be made to obtain a solution in the presence of existence difficulties. Most reconstructions shown in the literature are poor. In a comparative review, Zhang *et al.* (2002) summarize: "All the SFS algorithms produce generally poor results when given synthetic data . . . Results are even worse on real images, and . . . [r]esults on synthetic data are not generally predictive of results on real data." More recently, Tankus *et al.* (2005) showed good looking reconstructions of various body structures from endoscopic images, but cannot compare with veridical information. Prados and Faugeras show a good-looking reconstruction of a face, but cannot compare with veridical information (Prados and Faugeras (2005*a*); (2005*b*)). Durou *et al.* (2008*a*), in a recent comparative review, show some fair reconstructions on both synthetic and real data. However, on quite simple shapes methods still produce reconstructions with profound eccentricities.

These problems have driven a search for methods that do not require a reconstruction. Some local features of a shading field—*shading primitives*—are revealing because some geometric structures generate about the same shading pattern whatever the illumination. For example, a pit in a surface will always be dark; grooves and folds tend to appear as a thin, light band next to a thin, dark band; and the shading on a cylinder is usually either a dark band next to a light band, or a light band with dark band on either side. This idea originates with Koenderink and Doorn (1983), and is expounded in (Haddon and Forsyth 1997, Haddon and Forsyth 1998*b*, Han and Zhu 2005, Han and Zhu 2007). On a larger spatial scale, the family of shading patterns that can be produced by a particular object—the *illumination cone*—is smaller than one might expect (Basri and Jacobs 2003, Belhumeur and Kriegman 1998, Georghiades *et al.* 2001), allowing illumination invariant detection by detecting elements of such cones or by matching with an image distance that discounts changes in illumination (Chen *et al.* 2000, Jacobs *et al.* 1998).

PROBLEMS

2.1. We see a diffuse sphere centered at the origin, with radius one and albedo ρ, in an orthographic camera, looking down the z-axis. This sphere is illuminated by a distant point light source whose source direction is $(0, 0, 1)$. There is no other illumination. Show that the shading field in the camera is

$$\rho\sqrt{1 - x^2 - y^2}$$

2.2. What shapes can the shadow of a sphere take if it is cast on a plane and the source is a point source?

2.3. We have a square area source and a square occluder, both parallel to a plane. The source is the same size as the occluder, and they are vertically above one another with their centers aligned.
  (a) What is the shape of the umbra?
  (b) What is the shape of the outside boundary of the penumbra?

2.4. We have a square area source and a square occluder, both parallel to a plane. The edge length of the source is now twice that of the occluder, and they are vertically above one another with their centers aligned.
  (a) What is the shape of the umbra?
  (b) What is the shape of the outside boundary of the penumbra?

2.5. We have a square area source and a square occluder, both parallel to a plane. The edge length of the source is now half that of the occluder, and they are vertically above one another with their centers aligned.
  (a) What is the shape of the umbra?
  (b) What is the shape of the outside boundary of the penumbra?

2.6. A small sphere casts a shadow on a larger sphere. Describe the possible shadow boundaries that occur.

2.7. Explain why it is difficult to use shadow boundaries to infer shape, particularly if the shadow is cast onto a curved surface.

2.8. As in Figure 2.18, a small patch views an infinite plane at unit distance. The patch is sufficiently small that it reflects a trivial quantity of light onto the plane. The plane has radiosity $B(x, y) = 1 + \sin ax$. The patch and the plane are parallel to one another. We move the patch around parallel to the plane, and consider its radiosity at various points.
  (a) Show that if one translates the patch, its radiosity varies periodically with its position in x.
  (b) Fix the patch's center at $(0, 0)$; determine a *closed form* expression for the radiosity of the patch at this point as a function of a. You'll need a table of integrals for this (if you don't, you're entitled to feel very pleased with yourself).

2.9. If one looks across a large bay in the daytime, it is often hard to distinguish the mountains on the opposite side; near sunset, they are clearly visible. This phenomenon has to do with scattering of light by air—a large volume of air is actually a source. Explain what is happening. We have modeled air as a vacuum and asserted that no energy is lost along a straight line in a vacuum. Use your explanation to give an estimate of the kind of scales over which that model is acceptable.

2.10. Read the book *Colour and Light in Nature*, by Lynch and Livingstone, published by Cambridge University Press, 1995.

PROGRAMMING EXERCISES

2.11. An area source can be approximated as a grid of point sources. The weakness of this approximation is that the penumbra contains quantization errors, which can be quite offensive to the eye.

(a) Explain.

(b) Render this effect for a square source and a single occluder casting a shadow onto an infinite plane. For a fixed geometry, you should find that as the number of point sources goes up, the quantization error goes down.

(c) This approximation has the unpleasant property that it is possible to produce arbitrarily large quantization errors with any finite grid by changing the geometry. This is because there are configurations of source and occluder that produce large penumbrae. Use a square source and a single occluder, casting a shadow onto an infinite plane, to explain this effect.

2.12. Make a world of black objects and another of white objects (paper, glue, and spraypaint are useful here) and observe the effects of interreflections. Can you come up with a criterion that reliably tells, *from an image*, which is which? (If you can, publish it; the problem looks easy, but isn't.)

2.13. (This exercise requires some knowledge of numerical analysis.) Do the numerical integrals required to reproduce Figure 2.18. These integrals aren't particularly easy: if one uses coordinates on the infinite plane, the size of the domain is a nuisance; if one converts to coordinates on the view hemisphere of the patch, the frequency of the radiance becomes infinite at the boundary of the hemisphere. The best way to estimate these integrals is using a Monte Carlo method on the hemisphere. You should use importance sampling because the boundary contributes rather less to the integral than the top does.

2.14. Set up and solve the linear equations for an interreflection solution for the interior of a cube with a small square source in the center of the ceiling.

2.15. Implement a photometric stereo system.

(a) How accurate are its measurements (i.e., how well do they compare with known shape information)? Do interreflections affect the accuracy?

(b) How repeatable are its measurements (i.e., if you obtain another set of images, perhaps under different illuminants, and recover shape from those, how does the new shape compare with the old)?

(c) Compare the minimization approach to reconstruction with the integration approach; which is more accurate or more repeatable and why? Does this difference appear in experiment?

(d) One possible way to improve the integration approach is to obtain depths by integrating over many different paths and then average these depths (you need to be a little careful about constants here). Does this improve the accuracy or repeatability of the method?

CHAPTER 3

Color

The light receptors in cameras and in the eye respond more or less strongly to different wavelengths of light. Most cameras and most eyes have several different types of receptor, whose sensitivity to different wavelengths varies. Comparing the response of several types of sensor yields information about the distribution of energy with wavelength for the incoming light; this is color information. Color information can be used to identify specularities in images and to remove shadows. The color of an object seen in an image depends on how the object was lit, but there are algorithms that can correct for this effect.

3.1 HUMAN COLOR PERCEPTION

The light coming out of sources or reflected from surfaces has more or less energy at different wavelengths, depending on the processes that produced the light. This distribution of energy with wavelength is sometimes called a *spectral energy density*; Figure 3.1 shows spectral energy densities for sunlight measured under a variety of different conditions. The visual system responds to light in a range of wavelengths from approximately 400nm to approximately 700nm. Light containing energy at just one wavelength looks deeply colored (these colors are known as *spectral colors*). The colors seen at different wavelengths have a set of conventional names, which originate with Isaac Newton (the sequence from 700nm to 400nm goes Red Orange Yellow Green Blue Indigo Violet, or **R**ichard **o**f **Y**ork **g**ot **b**listers **i**n **V**enice, although indigo is now frowned upon as a name because people typically cannot distinguish indigo from blue or violet). If the intensity is relatively uniform across the wavelengths, the light will look white.

Different kinds of color receptor in the human eye respond more or less strongly to light at different wavelengths, producing a signal that is interpreted as color by the human vision system. The precise interpretation of a particular light is a complex function of context; illumination, memory, object identity, and emotion can all play a part. The simplest question is to understand which spectral energy densities produce the same response from people under simple viewing conditions (Section 3.1.1). This yields a simple, linear theory of color matching that is accurate and extremely useful for describing colors. We sketch the mechanisms underlying the transduction of color in Section 3.1.2.

3.1.1 Color Matching

The simplest case of color perception is obtained when only two colors are in view on a black background. In a typical experiment, a subject sees a colored light—the *test light*—in one half of a split field (Figure 3.2). The subject can then adjust a mixture of lights in the other half to get it to match. The adjustments involve changing the intensity of some fixed number of *primaries* in the mixture. In this

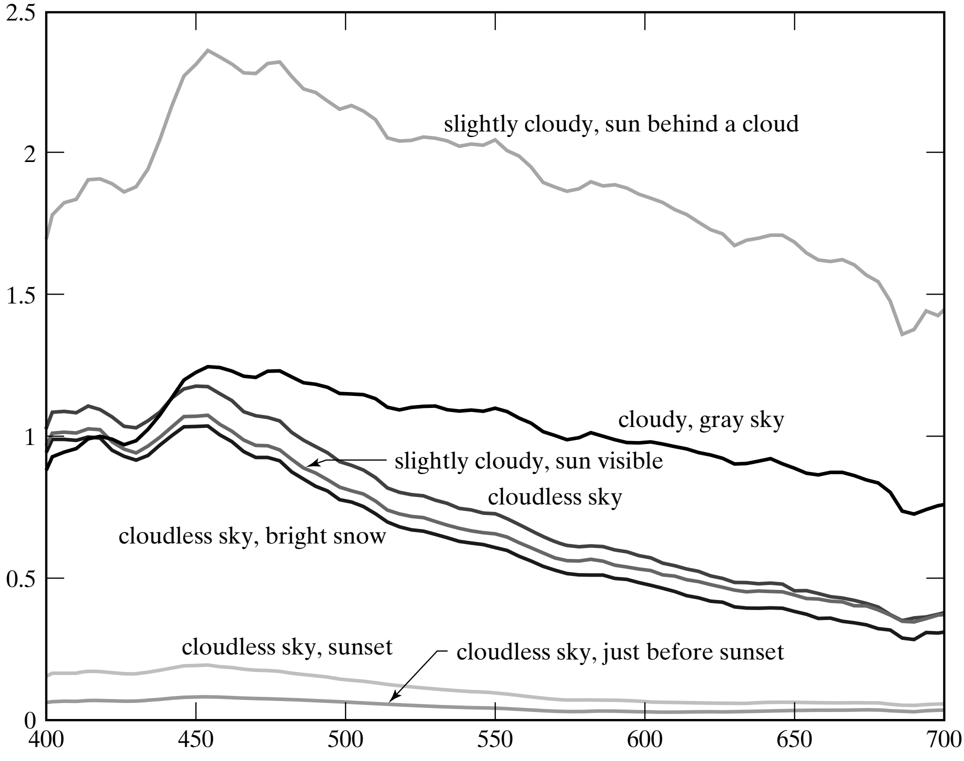

FIGURE 3.1: Daylight has different amounts of power at different wavelengths. These plots show the spectral energy density of daylight measured at different times of day and under different conditions. The figure plots relative power against wavelength for wavelengths from 400 nm to 700 nm for a series of seven different daylight measurements, made by Jussi Parkkinen and Pertti Silfsten, of daylight illuminating a sample of barium sulphate (which gives a high reflectance white surface). At the foot of the plot, we show the names used for spectral colors of the relevant wavelengths. Plot from data obtainable at `http://www.it.lut.fi/ip/research/color/database/database.html`.

form, a large number of lights may be required to obtain a match, but many different adjustments may yield a match.

Write T for the test light, an equals sign for a match, the weights—which are non-negative—as w_i, and the primaries P_i. A match can then be written in an algebraic form as

$$T = w_1 P_1 + w_2 P_2 + \dots,$$

meaning that test light T matches the particular mixture of primaries given by $(w_1, w_2, \dots)$. The situation is simplified if *subtractive matching* is allowed. In subtractive matching, the viewer can add some amount of some primaries to the *test light* instead of to the match. This can be written in algebraic form by allowing the weights in the expression above to be negative.

Under these conditions, most observers require only three primaries to match a test light. This phenomenon is known as the principle of *trichromacy*. However,